

The Collected Mathematical Papers of Arthur Cayley, Vol. VI

Pages 101–150 (Papers 396–402)

Transcription

396. On a Certain Envelope Depending on a Triangle Inscribed in a Circle

[From the Philosophical Magazine, vol. xxxiv. (1867), pp. 174–179.]

[Continued from p. 73.]

[Pp. 175—179.]

It is interesting to verify this; I write $Z = -X - Y$, the cubic function then assumes the form

$$(a, \beta, \gamma, \delta X, Y)^3,$$

where $(a, \beta, \gamma, \delta)$ have the values presently given.

The Hessian is

$$(2a\gamma - 2\beta^2)X^2 + (a\delta - \beta\gamma) \cdot 2XY + (2\beta\delta - 2\gamma^2)Y^2,$$

or writing $2XY = Z^2 - X^2 - Y^2$, this is

$$= (2a\gamma - 2\beta^2 - a\delta + \beta\gamma)X^2 + (2\beta\delta - 2\gamma^2 - a\delta - \beta\gamma)Y^2 + (a\delta - \beta\gamma)Z^2.$$

We find after some easy reductions,

$$\begin{aligned} J_0 &= Z^2 + L'^2, = -3(a+c)(ac+M), \\ I' &= -H' + 2K' + I' + F', = 3a(a+c)(b+c), \\ \gamma &= -K' + 2H' + J' + G', = 3b(a+c)(b+c), \\ \delta &= I' + J', = -3(b+c)(bc+M), \end{aligned}$$

and hence

$$a\delta - \beta\gamma = 81(a+c)(b+c)\{(ac+M)(bc+M) - ab(a+c)(b+c)\},$$

where the expression in $\{\}$ is

$$\begin{aligned} &= (ab + 2ac + bc)(ab + ac + 2bc) - ab(ab + ac + bc + c^2), \\ &= c\{ab(3a + 3b) + c(b + 2a)(a + 2b) - ab(a + b + c)\}, \\ &= 2c(a+b)(bc + ca + ab), \end{aligned}$$

and therefore

$$a\delta - \beta\gamma = -162(b+c)(c+a)(a+b)M;$$

the other coefficients may be similarly calculated, and omitting the merely numerical factor, we have

$$\text{Hess} = (b+c)(c+a)(a+b)M(aX^2 + bY^2 + cZ^2),$$

which is right.

I write next

$$\begin{aligned} HU + 3MU - \psi U &= X^2(-a^2x + I'Y - F'Z) \\ &\quad + Y^2(-b^2y + J'Z - G'X) + Z^2(-a^2z + K'X - H'Y), \end{aligned}$$

or writing $x = z - Y$, $y = z + X$, this is

$$HU + 3MU - \psi U = -4^2 z(aX^2 + bY^2 + cZ^2) \\ + X^2\{(I' + a^2)Y - F'Z\} + Y^2\{J'Z - (G' + b^2)X\} + Z^2\{K'X - H'Y\},$$

we may determine ψ , so that the cubic function of X, Y, Z contains the factor $aX^2 + bY^2 + cZ^2$; writing $Z = -X - Y$, then

$$\begin{aligned} &\text{Contains the factor } (a + c)X^2 + 2cXY + (b + c)Y^2 \\ &\text{Quotient is } -3(ac + M)X + 3(bc + M)Y. \end{aligned}$$

$$Z^2(\lambda I'' + F') + Z^2Y(-H' + 2K' + I' + \frac{1}{4}\psi - a^2) \\ + ZY^2(K' - 2H' - G' - J' - b^2 + \psi) + Y^3(-J' - H').$$

We have seen that

$$\begin{aligned} K' + F' &= -3(a + c)(ac + M), \\ J' + H' &= -3(b + c)(bc + M), \end{aligned}$$

whence the quotient is, as above stated,

$$= -3(ac + M)X + 3(bc + M)Y.$$

Comparing the coefficients of X^2Y , we have

$$\begin{aligned} a\psi &= -(-F' + 2K' + I' + I') + 3(a + c)(bc + M) - 6c(ac + M), \\ &= 9a(a + c)(b + c) + 3(a + c)(ab + ac + 2bc) - 6c(ab + 2ac + bc), \\ &= 12a(bc + ca + ab) = 12aM, \end{aligned}$$

that is $\psi = 12M$; and the same value would have been obtained by comparing the coefficients of XY^2 . Hence $HU + 3MU - \psi U$ divides by $aX^2 + bY^2 + cZ^2$, the quotient being

$$\begin{aligned} &= -12Mz - 3(ac + M)X + 3(bc + M)Y, \\ &= -12Mz - 3(ac + M)(y - z) + 3(bc + M)(z - x), \\ &= -3\{(bc + M)x + (ca + M)y + (ab + M)z\}, \end{aligned}$$

which is or, finally it is

$$= -3\{(bc + M)x + (ca + M)y + (ab + M)z\},$$

and we thus have

$$HU + 3MU - \psi U = -3(aX^2 + bY^2 + cZ^2) \times \{(bc + M)x + (ca + M)y + (ab + M)z\},$$

so that the three inflexions are the intersections of the cubic curve by the line

$$(bc + M)x + (ca + M)y + (ab + M)z = 0.$$

It may be noticed, that if we write

$$\begin{aligned} x + y + z &= u, \\ bex + cay + abz &= -Mu, \\ ax + by + cz &= v, \end{aligned}$$

then x, y, z will be as

$$(b - c)\{(2M - bc)u - av\} : (c - a)\{(2M - ca)u - bv\} : (a - b)\{(2M - ab)u - cv\},$$

and substituting these values in the equation

$$ax(y-z)^2 + by(z-x)^2 + cz(x-y)^2 = 0$$

of the cubic, we have a cubic equation for the ratio $(u : v)$; and thence the values (x, y, z) for the coordinates of the inflexions.

It may be added, that we have

$$12MU = -3(aX^2 + bY^2 + cZ^2)\{(bc + M)x + (ca + M)y + (ab + M)z\} \\ + \{X^2(FY - F'Z) + Y^2(J'Z - G'X) + Z^2(K'X - H'Y)\} = 0,$$

which is the equation of the cubic expressed in the canonical form.

[Pp. 175—179.]

Effecting the process indicated p. 73, but writing for greater convenience (x, y, z) in place of (X, Y, Z) , so that the substitution to be made is

$$(a, b, c, f, g, h, i, j, k, l) = (6x, 6y, 6z, -2z, -2x, -2z, -2x, -2y, x + y + z),$$

respectively (where I have corrected a misprint in the formula as originally given) I find the equation of the envelope to be

$$4y^2(y-z)^2a^4 + 4z^2x(z-x)^2b^4 + 4x^2y(x-y)^2c^4 \\ + 4zx(z^2 + x^2 + 3yz - 2zx + 5xy)b^2c \\ + 4xy(x^2 + y^2 + 3zx - 2xy + 5yz)c^2a \\ + 4yz(y^2 + z^2 + 3xy - 2yz + 5zx)a^2b \\ + 4xy(x^2 + y^2 + 3yz - 2xy + 5zx)bc^2 \\ + 4y^2(y^2 + z^2 + 3zx - 2yz + 5xy)ca^2 \\ + 4zx(z^2 + x^2 + 3xy - 2zx + 5yz)ab^2 \\ + x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2)b^2c^2 \\ + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2)c^2a^2 \\ + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2)a^2b^2 \\ + 2yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx)a^2bc \\ + 2zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy)b^2ca \\ + 2xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz)c^2ab = 0.$$

The function on the left-hand side is the quotient by $-48(a+b+c)^2$ of the sextic function, Table 67 of my third Memoir on Quantics, [144]; the foregoing quotient was calculated without using the coefficient of the term in $a^2b^2c^2$ (viz.?"") of the table, but by way of verification, I calculated from the table the term in question, and found it to be

$$(x+y+z)^4 - 2(x+y+z)^2(yz+zx+xy) \\ + 296(x+y+z)xyz - 8(y^2z^2 + z^2x^2 + x^2y^2),$$

and this should consequently be equal to the coefficient of $a^2b^2c^2$ in the product of $(a+b+c)^2$ into the foregoing quartic function of (a, b, c) that is, it should be

$$= x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2) \\ + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2) \\ + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2) \\ + 4yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx) \\ + 4zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy) \\ + 4xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz),$$

which is accordingly found to be the case.

397. Specimen Table $M \equiv a^\alpha b^\beta \pmod{N}$ for Any Prime or Composite Modulus

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 95—96 and plate.]

If N be a prime number, and a one of its primitive roots, then any number M prime to N , or what is the same thing, any number in the series $1, 2, \dots, N-1$, may be exhibited in the form $M = a^\alpha \pmod{N}$; where α is said to be the *index* of M in regard to the particular root a .

Jacobi's *Canon Arithmeticus* (Berlin, 1839), contains a series of tables, giving the indices of the numbers $1, 2, 3, \dots, N-1$ for every prime number N less than 1000, and giving conversely for each such prime number the numbers M which correspond to the indices $\alpha = 1, 2, \dots, (N-1)$ (Tabulæ Numerorum ad Indices datos pertinentium et Indicium Numero dato correspondentium).

A similar theory applies, it is well known, to the composite numbers; the only difference is, that in order to exhibit for a given composite number N the different numbers less than N and prime to it, we require not a single root a , but two or more roots $a, b, \dots$ and that in terms of these we have $M = a^\alpha b^\beta \dots \pmod{N}$.

For each root a there is an index A (or say the Indicator of the root), such that $a^A = 1 \pmod{N}$, A being the least index for which this equation is satisfied; and the indices $\alpha, \beta, \dots$ extend from 1 to $A, B, \dots$ respectively; the number of different combinations or the product $AB \dots$, being precisely equal to $\phi(N)$, the number of integers less than N and prime to it.

The least common multiple of $A, B, \dots$, is termed the Maximum Indicator, and representing it by I , then for any number M not prime to N , we have $M^I = 1 \pmod{N}$, a theorem made use of by Cauchy for the solution of indeterminate equations of the first order.

Thus $N = 20$, the roots may be taken to be 3, 11; the corresponding exponents are 4, 2 (viz. $3^4 \equiv 1 \pmod{20}$, $11^2 \equiv 1 \pmod{20}$), and the product of these is 8, the number of integers less than 20 and prime to it; the series [go to p. 86]

[Plate pages 84–85. Tables of indices for moduli $N = 1$ to 50. The tables occupy two full pages with the following arrangement:]

Table arrangement. For each modulus N from 1 to 50, the table presents five heading lines:

1. The modulus N ;
2. The root or roots belonging to each number N ;
3. The indicators of these roots;
4. The maximum indicator;
5. The number $\phi(N)$.

The remaining lines contain the index or indices of each of the $\phi(N)$ numbers M less than N and prime to it, the number corresponding to such index or indices being placed outside in the same horizontal line.

Specimen: $N = 30$.

30 has the roots 7, 11, indices 4, 2 respectively; the Maximum Indicator is 4, and the number of integers less than 30 and prime to it is 8; taking any such number, say 17, the indices are 1, 1, that is, we have $17 = 7^1 \cdot 11^1 \pmod{30}$.

[From p. 83]

The foregoing corresponds to the *Tabulæ Indicum Numero dato correspondentium* of Jacobi; on account of multiplicity of roots there does not appear to be any mode of forming a single table corresponding to the *Tabulæ Numerorum ad Indices datos pertinentium*; and there would be no adequate advantage in forming for each number N a separate table in some such form as

$N = 20$		
Roots	3	11
Nos.		
1	0	0
3	1	0
7	1	1
9	2	0
11	0	1
13	2	1
17	3	1
19	3	0

which I have written down in the form of a table of single entry; for although (whenever, as in the present case, the number of roots is only two) it might have been better exhibited as a table of double entry, when the number of roots is three or more it could not of course be exhibited as a table of corresponding multiple entry.

398. On a Certain Sextic Developable, and Sextic Surface Connected Therewith

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 129—142 and 373—376.]

I PROPOSE to consider [first] the sextic developable derived from a quartic equation, viz. taking this to be $(a, b, c, d, et, 1)^4 = 0$, where (a, b, c, d, e) are any linear functions of the coordinates (x, y, z, w) , the equation of the developable in question is

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

I have already, in the paper “On a Special Sextic Developable,” *Quarterly Journal of Mathematics*, vol. VII. (1866), pp. 105—113, [373], considered a particular case of this surface, viz. that in which c was $= 0$, the geometrical peculiarity of which is that the cuspidal edge is there an excubo-quartic curve (of a special form, having two stationary tangents), whereas in the general case here considered it is a sextic curve.

There was analytically the convenience that the linear functions being only the four functions a, b, d, e , these could be themselves taken as coordinates, whereas in the present case we have the five linear functions a, b, c, d, e .

The developable $(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0$ is a sextic developable having for its cuspidal curve the sextic curve

$$\begin{aligned} ae - 4bd + 3c^2 &= 0, \\ ace - ad^2 - b^2e + 2bcd - c^3 &= 0, \end{aligned}$$

(say $I = 0, J = 0$, as usual), and having besides a nodal curve the equations of which may be written

$$6(ac - b^2) : 3(ad - bc) : ae + 2bd - 3c^2 : 3(be - cd) : 6(ce - d^2) = a : b : c : d : e.$$

viz. these equations are really equivalent to two equations, and they represent a curve of the fourth order which is an excubo-quartic.

We may in fact find the equations of the nodal curve by assuming $(a, b, c, d, et, 1)^4$ to be a perfect square, say to avoid fractions that it is $= 3(\alpha t^2 + 2\beta t + \gamma)^2$, then we have

$$a : b : c : d : e = 3\alpha^2 : 3\alpha\beta : \alpha\gamma + 2\beta^2 : \beta\gamma : 3\gamma^2,$$

which equations as involving the two arbitrary parameters $\alpha : \beta : \gamma$, give two equations between (a, b, c, d, e) , and we may at once by means of them verify the above-mentioned equations of the nodal curve.

It also hereby appears that the nodal curve is as stated an excubo-quartic curve; viz. we have between a, b, c, d, e a single linear relation, that is a quadric relation between α, β, γ , and this equation may be satisfied identically by taking for α, β, γ properly determined quadric functions of a variable parameter θ ; whence a, b, c, d, e are proportional to quartic functions of the variable parameter θ , or the curve is an excubo-quartic.

The equations of the nodal curve may be presented under a somewhat different form; viz. the cubi-covariant of $(a, b, c, d, et, 1)^4 = 0$ being

$$\begin{aligned} &(-a^2d + 3abc - 2b^3 - a^2e - 2abd + 9ac^2 - 6b^2c \\ &\quad - abe + 6acd - 6bc^2 + 3ad^2 - b^2e \\ &\quad + ade + 2bd^2 - 3bce \\ &\quad + ae^2 + 2bde - 9c^2e + 6cd^2 \\ &\quad + be^2 - 3cde + 2d^3t, 1)^3 = 0, \end{aligned}$$

say this function, multiplied by 6 to avoid fractions, is

$$(a, b, c, d, e, f, gt, 1)^3,$$

that is

$$\begin{aligned} a &= 6(-a^2d + 3abc - 2b^3), \\ b &= (-a^2e - 2abd + 9ac^2 - 6b^2c), \\ c &= 2(-abe + 6acd - 6bc^2), \\ d &= 3(+ad^2 - b^2e), \\ e &= 2(+ade + 2bd^2 - 3bce), \\ f &= (+ae^2 + 2bde - 9c^2e + 6cd^2), \\ g &= 6(+be^2 - 3cde + 2d^3), \end{aligned}$$

then the equations of the nodal curve may be written:

$$a = 0, \quad b = 0, \quad c = 0, \quad d = 0, \quad e = 0, \quad f = 0, \quad g = 0.$$

It may be mentioned that we have identically

$$\begin{aligned} ae - 4bd + 3c^2 &= 0, \\ af - 3be + 2cd &= 0, \\ ag - 9ce + 8d^2 &= 0, \\ bg - 3cf + 2de &= 0, \\ bf - 4ce + 3d^2 &= eg - 3df + 2e^2 = 0. \end{aligned}$$

and moreover

$$ag - 6bf + 15ce - 10d^2 = -6(bf - 4ce + 3d^2) = +6(I^2 - 27J^2),$$

so that the equation of the developable may be written in the form

$$ag - 6bf + 15ce - 10d^2 = 0,$$

or in the more simple form

$$bf - 4ce + 3d^2 = 0,$$

each of which puts in evidence the nodal curve on the surface.

The nodal and cuspidal curves meet in the points

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e},$$

being, as it is easy to show, a system of four points.

The four points in question form a tetrahedron, the equations of the faces of which may be taken to be $x = 0$, $y = 0$, $z = 0$, $w = 0$; and the equation of the surface may be expressed in this system of quadriplanar coordinates.

We introduce these coordinates *ab initio*, by taking the quartic function of t to be

$$(a, b, c, d, e, 1)^4 = x'(t + \alpha)^4 + y'(t + \beta)^4 + z'(t + \gamma)^4 + w'(t + \delta)^4,$$

that is, by writing

$$\begin{aligned} a &= x' + y' + z' + w', \\ b &= \alpha x' + \beta y' + \gamma z' + \delta w', \\ c &= \alpha^2 x' + \beta^2 y' + \gamma^2 z' + \delta^2 w', \\ d &= \alpha^3 x' + \beta^3 y' + \gamma^3 z' + \delta^3 w', \\ e &= \alpha^4 x' + \beta^4 y' + \gamma^4 z' + \delta^4 w'. \end{aligned}$$

Observe that (t_1, t_2, t_3, t_4) being any constant quantities, we thence have

$$\begin{aligned} e - d\Sigma t_1 + c\Sigma t_1 t_2 - b\Sigma t_1 t_2 t_3 + at_1 t_2 t_3 t_4 \\ = x'(\alpha - t_1)(\alpha - t_2)(\alpha - t_3)(\alpha - t_4) \\ + y'(\beta - t_1)(\beta - t_2)(\beta - t_3)(\beta - t_4) \\ + z'(\gamma - t_1)(\gamma - t_2)(\gamma - t_3)(\gamma - t_4) \\ + w'(\delta - t_1)(\delta - t_2)(\delta - t_3)(\delta - t_4), \end{aligned}$$

and thence in particular

$$e - d\Sigma\alpha + c\Sigma\alpha\beta - b\Sigma\alpha\beta\gamma + a\alpha\beta\gamma\delta = 0,$$

viz. this is the linear relation which subsists identically between (a, b, c, d, e) , the five linear functions of the coordinates (x, y, z, w) .

Starting from the above values of (a, b, c, d, e) , we find without difficulty

$$I = (\alpha - \beta)^2 x' y' + (\alpha - \gamma)^2 x' z' + (\alpha - \delta)^2 x' w' \\ + (\beta - \gamma)^2 y' z' + (\beta - \delta)^2 y' w' + (\gamma - \delta)^2 z' w',$$

$$J = (\alpha - \beta)^2 (\alpha - \gamma)^2 (\beta - \gamma)^2 x' y' z' + (\alpha - \beta)^2 (\alpha - \delta)^2 (\beta - \delta)^2 x' y' w' \\ + (\alpha - \gamma)^2 (\gamma - \delta)^2 (\alpha - \delta)^2 x' z' w' + (\beta - \gamma)^2 (\gamma - \delta)^2 (\beta - \delta)^2 y' z' w',$$

but we thus see the convenience of introducing constant multipliers into the expressions of the four coordinates respectively, viz. writing

$$X = (\beta\gamma\delta)x', \quad Y = (\gamma\delta\alpha)y', \quad Z = (\delta\alpha\beta)z', \quad W = (\alpha\beta\gamma)w',$$

where for shortness $(\beta\gamma\delta) = (\beta - \gamma)(\gamma - \delta)(\delta - \beta)$, &c., or what is the same thing, taking the quartic to be

$$(a, b, c, d, et, 1)^4 = X'(\beta\gamma\delta)(t + \alpha)^4 + Y'(\gamma\delta\alpha)(t + \beta)^4 + Z'(\delta\alpha\beta)(t + \gamma)^4 + W'(\alpha\beta\gamma)(t + \delta)^4,$$

we find

$$J = (\lambda'\mu'\nu')^2 (x'y'z' + y'z'w' + z'x'w' + x'y'w'), \\ I = (\lambda'\mu'\nu') \{ \lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y') \},$$

where for shortness

$$\lambda' = (\alpha - \delta)^2 (\beta - \gamma)^2, \\ \mu' = (\beta - \delta)^2 (\gamma - \alpha)^2, \\ \nu' = (\gamma - \delta)^2 (\alpha - \beta)^2, \\ \nabla(\lambda') = (\alpha - \delta)(\beta - \gamma), \\ \nabla(\mu') = (\beta - \delta)(\gamma - \alpha), \\ \nabla(\nu') = (\gamma - \delta)(\alpha - \beta), \\ \nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0,$$

and the equation of the developable is thus

$$\{ \lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y') \}^3 - 27\lambda'\mu'\nu'(x'y'z' + y'z'w' + z'x'w' + x'y'w')^2 = 0.$$

Observe that $J = 0$ is a cubic surface passing through each edge of the tetrahedron, and having at each summit a conical point; $I = 0$ is a quadric surface passing through each summit of the tetrahedron, and at each of these points the tangent plane of the quadric surface touches the tangent cone of the cubic surface: to show this it is only necessary to observe that at the point $(x' = 0, y' = 0, z' = 0)$ the tangent cone is $y'z' + z'x' + x'y' = 0$, and the tangent plane is $\lambda'x' + \mu'y' + \nu'z' = 0$, and that these touch in virtue of the above-mentioned relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$.

It follows that on the curve of intersection, or cuspidal edge of the developable, each of the summits is a cuspidal or stationary point, that is, the cuspidal curve has four stationary points; this agrees with the character of the curve as given, Salmon "On the Classification of Curves of Double Curvature," Camb. and Dubl. Math. Jour. vol. V. (1850), p. 39, viz. the character is there given $a = 6, m = 6, n = 4, r = 6, g = 3, h = 6, \alpha = 0, \beta = 4, \kappa = 4, \gamma = 6, (\beta = 4$, that is, there are as stated 4 stationary points).

To find the equations of the nodal curve, instead of transforming the equations as given in terms of (a, b, c, d, e) , it is better to deduce these from the equation of the surface; viz. if there is a nodal curve, we must have

$$\partial_{x'}I : \partial_{y'}I : \partial_{z'}I : \partial_{w'}I = \partial_{x'}J : \partial_{y'}J : \partial_{z'}J : \partial_{w'}J.$$

Writing these under the form $\partial_{x'}I + \theta' \partial_{x'}J = 0$, &c., where θ' is regarded as an arbitrary parameter¹, we have

$$\begin{aligned} \lambda'w' + \mu'z' + \nu'y' + \theta'(y'z' + y'w' + z'w') &= 0, \\ \lambda'z' + \mu'w' + \nu'x' + \theta'(z'x' + z'w' + x'w') &= 0, \\ \lambda'y' + \mu'x' + \nu'w' + \theta'(x'y' + x'w' + y'w') &= 0, \\ \lambda'x' + \mu'y' + \nu'z' + \theta'(y'z' + z'x' + x'y') &= 0, \end{aligned}$$

which equations (eliminating θ') must be equivalent to two equations only.

I remark that the first three equations may be regarded as a set of linear equations in $1, w', \theta', \theta'w'$; and determining from them the ratios of these quantities, we have, suppose,

$$1 : -w' : \theta' : -\theta'w' = A : B : C : D,$$

where

$$A = \begin{vmatrix} \lambda' & \nu' & y' \\ \mu' & \lambda' & z' + x' \\ \nu' & \mu' & x' + y' \end{vmatrix}, \quad B = \begin{vmatrix} y'z' & y' + z' & \nu'z' + \mu'y' \\ z'x' & z' + x' & \lambda'z' + \nu'x' \\ x'y' & x' + y' & \mu'x' + \lambda'y' \end{vmatrix}, \quad \text{etc.}$$

¹The value of θ' is in fact $= \frac{18J}{I}$, that is, instead of the four equations involving an arbitrary parameter θ' , we have really four determinate equations.

We have thence $AD - BC = 0$ and (substituting in the fourth equation) $A(\lambda'x' + \mu'y' + \nu'z') + C(y'z' + z'x' + x'y') = 0$; each of these equations must contain the equation of the cone having $(x' = 0, y' = 0, z' = 0)$ for its vertex, and passing through the nodal curve.

The two equations are of the orders 6 and 4 respectively; and as the curve is a quartic curve passing through the vertex in question, the equation of the cone is of the order 3.

I have not effected the reduction of the sextic equation, but for the quartic equation, substituting for A, C their values, this is

$$\begin{aligned} & -(\lambda'w' + \mu'y' + \nu'z')\{\lambda'x'^2(y' - z') + \mu'y'^2(z' - x') + \nu'z'^2(x' - y')\} \\ & + (y'z' + z'x' + x'y')\{\lambda'x'(y'^2 - z'^2) + \mu'y'(z'^2 - x'^2) + \nu'z'(x'^2 - y'^2)\} \\ & + \{\mu'\nu'(y' - z') + \nu'\lambda'(z' - x') + \lambda'\mu'(x' - y')\}(x' + y' + z') = 0, \end{aligned}$$

which is easily reduced to

$$\begin{aligned} & \lambda'x'(-x'^2 + y'z' + z'x' + x'y')(y' - z') \\ & + \mu'y'(-y'^2 + y'z' + z'x' + x'y')(z' - x') \\ & + \nu'z'(-z'^2 + y'z' + z'x' + x'y')(x' - y') \\ & + \mu'\nu'\{(x' + y' + z')(y'z' + z'x' - x'y') + x'y'z'\}(y' - z') \\ & + \nu'\lambda'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(z' - x') \\ & + \lambda'\mu'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(x' - y') = 0; \end{aligned}$$

and I have found that this is transformable into

$$\begin{aligned} & 2\{x'^2\nabla(\lambda') + y'^2\nabla(\mu') + z'^2\nabla(\nu')\} \times [x'^2\nabla(\lambda')(\mu'y'^2 - \nu'z'^2) \cdot 2x'\nabla(\mu')(\nu'z'^2 - \lambda'x'^2) \cdot x'y'\nabla(\nu')(\lambda'x'^2 - \mu'y'^2) \\ & - x'^2y'^2z'^2\{\nabla(\mu') - \nabla(\nu')\}\{\nabla(\nu') - \nabla(\lambda')\}\{\nabla(\lambda') - \nabla(\mu')\}] = 0, \end{aligned}$$

viz. the two functions are equivalent in virtue of the relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$, or, what is the same thing, they only differ by a function $(x', y', z')^4$ into the evanescent factor $\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu'$.

The function in $\{\}$ equated to zero is therefore the equation of the cubic cone.

I do not stop to give the steps of the investigation in the above form, as the investigation may be very much simplified as follows: by linear combinations of the four equations in x', y', z', w', θ' , we deduce

$$\begin{aligned} & (\lambda' - \mu' - \nu')(x' + w' - y' - z') + 2\theta'(y'z' - x'w') = 0, \\ & (-\lambda' + \mu' - \nu')(y' + w' - z' - x') + 2\theta'(z'x' - y'w') = 0, \\ & (-\lambda' - \mu' + \nu')(z' + w' - x' - y') + 2\theta'(x'y' - z'w') = 0, \\ & (\lambda' + \mu' + \nu')(x' + y' + z' + w') + 2\theta'(y'z' + z'x' + x'y' + x'w' + y'w' + z'w') = 0. \end{aligned}$$

Hence writing

$$\begin{aligned} \lambda &= \lambda' - \mu' - \nu', & x &= w' + x' - y' - z', \\ \mu &= -\lambda' + \mu' - \nu', & y &= w' - x' + y' - z', \\ \nu &= -\lambda' - \mu' + \nu', & z &= w' - x' - y' + z', \\ & & w &= w' + x' + y' + z', \end{aligned}$$

we find

$$\begin{aligned}\lambda^2 &= \lambda'^2 - \mu'^2 - \nu'^2 + 2\mu'\nu', \\ \mu^2 &= -\lambda'^2 + \mu'^2 - \nu'^2 + 2\nu'\lambda', \\ \nu^2 &= -\lambda'^2 - \mu'^2 + \nu'^2 + 2\lambda'\mu',\end{aligned}$$

and thence

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = -(\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu')^2 = 0,$$

that is

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = 0, \quad \frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

the relation which connects the new constants λ, μ, ν .

Moreover

$$\begin{aligned}yz - xw &= 4(y'z' - x'w'), \\ zx - yw &= 4(z'x' - y'w'), \\ xy - zw &= 4(x'y' - z'w'), \\ 2w^2 - x^2 - y^2 - z^2 &= 8(y'z' + z'x' + x'y' + x'w' + y'w' + z'w'),\end{aligned}$$

and writing for greater convenience $\theta = -\frac{\theta'}{2}$, the equations are transformed into

$$\begin{aligned}\theta\lambda x &= vw - yz, \\ \theta\mu y &= yw - zx, \\ \theta\nu z &= zw - xy, \\ 2\theta(\lambda + \mu + \nu)w &= 3w^2 - x^2 - y^2 - z^2,\end{aligned}$$

where

$$\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

viz. these equations, eliminating θ , give the equations of the nodal curve.

From the first three equations eliminating θ , we deduce

$$\begin{aligned}yzw(\mu - \nu) &= x(\mu y^2 - \nu z^2), \\ zxw(\nu - \lambda) &= y(\nu z^2 - \lambda x^2), \\ xyw(\lambda - \mu) &= z(\lambda x^2 - \mu y^2),\end{aligned}$$

or, as these equations may be written,

$$\frac{x(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z(\lambda x^2 - \mu y^2)}{\lambda - \mu},$$

which equations, from the mode in which they are obtained, are it is clear equivalent to two equations only.

Using the fourth equation, and eliminating by substituting therein for $\theta\lambda$, $\theta\mu$, $\theta\nu$ their values from the first three equations, we find

$$2w^2 (3w^2 - x^2 - y^2 - z^2) = \frac{3w^2 - x^2 - y^2 - z^2}{\lambda} + \frac{3w^2 - x^2 - y^2 - z^2}{\mu} + \frac{3w^2 - x^2 - y^2 - z^2}{\nu}$$

that is

$$3w^2 + x^2 + y^2 + z^2 = 2w \left(\frac{x^2}{\lambda} + \frac{y^2}{\mu} + \frac{z^2}{\nu} \right),$$

or, what is the same thing,

$$xyz(3w^2 - x^2 + y^2 + z^2) - 2w(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

we have to show that this is in fact included in the former system, for then the four equations with θ eliminated will it is clear give two equations only.

Observe that the former system may be written

$$\frac{x(\mu y^2 - \nu z^2)}{(\mu - \nu)yz} = \frac{y(\nu z^2 - \lambda x^2)}{(\nu - \lambda)zx} = \frac{z(\lambda x^2 - \mu y^2)}{(\lambda - \mu)xy},$$

$$\begin{aligned} & (\mu - \nu)yz \cdot x(\mu y^2 - \nu z^2) + (\nu - \lambda)zx \cdot y(\nu z^2 - \lambda x^2) + (\lambda - \mu)xy \cdot z(\lambda x^2 - \mu y^2) \\ &= \mu\nu yz(\mu - \nu)(y^2 - z^2) + \nu\lambda zx(\nu - \lambda)(z^2 - x^2) + \lambda\mu xy(\lambda - \mu)(x^2 - y^2) \\ &+ xyz\{\lambda(\mu - \nu)(\mu y^2 + \nu z^2) + \mu(\nu - \lambda)(\nu z^2 + \lambda x^2) + \nu(\lambda - \mu)(\lambda x^2 + \mu y^2)\}. \end{aligned}$$

[The paper continues on pp. 373—376 with further development of the theory of the sextic developable and its connection with the sextic surface. The investigation involves detailed analysis of the nodal curve, the cuspidal edge, and their singularities, together with the coordinate transformations and algebraic relations connecting the different forms of the surface equations.]

399. On the Cubical Divergent Parabolas

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 185—190.]

[From p. 101.]

[Pp. 185—190.]

The cubical divergent parabolas are the parabolas $y^2 = x^3 + ax + b$, where the parameters a, b are such that the discriminant $4a^3 + 27b^2$ is $= 0$; viz. these are curves for which the line at infinity meets the curve in three distinct points, but the line at infinity is a tangent, or the curve has a cusp at infinity.

The several forms may be distinguished according to the nature of the roots of the equation $4a^3 + 27b^2 = 0$ considered as a cubic equation in $a : b$; or what is the same thing, according to the nature of the roots of the equation $z^3 - 3z + 2 = 0$ (where for convenience a, b have been replaced by $3z, -2$ respectively), viz. the roots are $z = 1, 1, -2$; and the forms correspond to the two cases $z = 1$ and $z = -2$.

For $z = 1$, that is $4a^3 + 27b^2 = 0, a \neq 0$, the equation is $y^2 = x^3 - 3x + 2$, that is $y^2 = (x-1)^2(x+2)$, which has a crunode at infinity; or writing $x = \frac{1}{\xi}, y = \frac{\eta}{\xi^2}$, the equation becomes $\eta^2 = (1-\xi)^2(1+2\xi)$, which has a crunode at $\xi = 0, \eta = 0$, viz. the origin.

For $z = -2$, that is $a = 0, b = 0$, the equation is $y^2 = x^3$, which has a cusp at infinity; or in the like transformed equation $\eta^2 = \xi^3$, which has a cusp at the origin.

The form $y^2 = x^3 - 3x + 2$, that is $y^2 = (x-1)^2(x+2)$, has a crunode at infinity; the tangent at the crunode is the line at infinity, and the two branches are each touched by the line at infinity. The form $y^2 = x^3$ has a cusp at infinity, the tangent at the cusp being the line at infinity.

It is convenient to exhibit the different species in a tabular form, indicating the nature of the singularities (finite and at infinity), and the sign of the discriminant $\Delta = -(4a^3 + 27b^2)$ in the different cases.

Equation	Singularities	Δ
$y^2 = x^3 + 3x - 2$	simplex acnodal, R without the curve	—
$y^2 = x^3$	simplex cuspidal, R at the cusp	0
$y^2 = x^3 - 3x + 2$	simplex crunodal, R at the node	+
$y^2 = x^3 + ax + b$ ($4a^3 + 27b^2 \neq 0$)	simplex non-singular	$\neq 0$

The species may be further subdivided according to the position of the point R (the intersection of the three inflexions) relative to the curve and its singularities. For the simplex non-singular form, we have according as R is within or without the curve, the campaniform or serpentine forms; and for the singular forms, corresponding subdivisions according as R coincides with or is distinct from the singularity, and in the latter case according as it lies within or without the curve.

400. On the Cubic Curves Inscribed in a Given Pencil of Six Lines

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 211—230.]

[From p. 105.]

[Pp. 211—230.]

I consider a system of six lines forming a pencil, that is, meeting in a point, and I seek to investigate the general equation of a cubic curve inscribed in the six lines; viz. the six lines are to be tangents to the cubic curve. It is to be observed that the six lines may be any six lines of a pencil; there is no restriction whatever as to the particular positions of the lines. The problem is: given six tangents of a cubic, to find the equation of the cubic; or what is the same thing, to find the equation of the curve having the six given lines for tangents.

Taking the centre of the pencil for the point $(x = 0, y = 0)$, and taking $z = 0$ for the equation of the line at infinity, the equations of the six tangents may be written

$$z = 0, \quad y - m_1x = 0, \quad y - m_2x = 0, \quad \dots, \quad y - m_5x = 0,$$

where m_1, m_2, m_3, m_4, m_5 are the slopes of the five finite tangents respectively. The general equation of a cubic curve is

$$(a, b, c, d, f, g, h, l, m, nx, y, z)^3 = 0,$$

or what is the same thing,

$$ax^3 + by^3 + cz^3 + 3dy^2z + 3ez^2x + 3fz^2y + 3gxz^2 + 3hyz^2 + 3lx^2y + 3mxy^2 + 6nxyz = 0.$$

The condition that the line $y - mx = 0$ may be a tangent to this cubic is obtained by substituting $y = mx$ in the equation of the cubic, and expressing that the resulting equation in $x : z$ has equal roots; viz. substituting $y = mx$, we have

$$(a + 3lm + 3m^2 + bm^3)x^3 + 3(n + fm + em^2 + hm^3 + gm^2 + dm^3)xz^2 + (c + 3gz + 3hz + dz)z^3 = 0,$$

or more simply

$$(a, b, c, d, e, f, g, h, l, m, n1, m, 0)^3x^3 + \dots = 0.$$

The condition of equal roots gives a sextic equation in m , which must be satisfied identically for $m = m_1, m_2, m_3, m_4, m_5$ and $m = \infty$ (corresponding to the tangent $z = 0$). We hence obtain the required conditions on the coefficients.

The investigation is conducted by means of the invariants of the cubic. Writing the cubic in the canonical form, and making use of the known relations between the invariants and the coefficients, the conditions that six given lines of a pencil shall be tangents to the cubic may be expressed in a very elegant form.

Let the six tangents be denoted by their equations $l_1 = 0, l_2 = 0, \dots, l_6 = 0$, where $l_i = y - m_i x$ for $i = 1, \dots, 5$, and $l_6 = z$. Then the equation of any cubic having these six lines for tangents may be written in the form

$$\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6} = 0,$$

where the signs of the radicals are properly determined.

Or more definitely, if we write

$$\sqrt{l_1} \pm \sqrt{l_2} \pm \sqrt{l_3} \pm \sqrt{l_4} \pm \sqrt{l_5} \pm \sqrt{l_6} = 0,$$

the different systems of signs give the different cubics which have the six lines for tangents. It is known that there are in general ten such cubics; the different systems of signs must be such that the product of all the terms is positive, and the number of independent systems is ten.

The equation may also be written in the rationalised form. We have

$$(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6})(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} - \sqrt{l_6}) \cdots = 0,$$

where the product contains $2^5 = 32$ factors, but these combine in pairs to give 16 different cubics, and of these only 10 are proper cubics, the remaining 6 being degenerate forms.

For the complete discussion it is necessary to introduce the fundamental invariants S and T of the cubic. These are connected with the coefficients by the relations

$$S = \frac{1}{3 \cdot 12^4}(abc + \dots),$$

$$T = \frac{1}{27 \cdot 12^6}(a^2b^2c^2 + \dots),$$

and the discriminant is $\Delta = 64S^3 - T^2$.

For a cubic having six given tangents which form a pencil, the invariants S and T satisfy certain relations which may be obtained as follows. Let the six lines be given, and let the cubic be expressed in terms of the parameters which determine the position of the point of contact on each tangent; then the condition that the six points of contact shall lie on a cubic curve gives the required relation between S and T .

The detailed calculation is as follows. Let the cubic be written in the form

$$c[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2 y^2 = 0,$$

where $(a, h, k, bx, y)^2 = ax^2 + 2hxy + by^2$, and $\Delta = hk - ab$, $V = hk - h^2$, $\Theta = ah - bk$, $R = -81ab$.

The discriminant of this form, say Q_{10} , is a function of c^2D , V , $c\Theta$, cR , where D is the discriminant of the quadratic $(a, h, k, bx, y)^2$. The expression is found to be

$$\begin{aligned} Q_{10} = & c^{10}\{D^3\} + c^9\{-99RD^2 - 400VD + 2304V\Theta D + 8640\Theta^2\} \\ & + c^8\{12800R^2V\Theta + 82944V^3\Theta^2 - 4608V^2D\} \\ & + c^7\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\}. \end{aligned}$$

The foregoing values of S and T give

$$T^2 - 64S^3 = c^4\{D^3 + c(8RD^2 - 24V\Theta D - 64\Theta^3) + c^2(16R^2 + 96RV\Theta - 48V^2\Theta^2 + 16V^3D) + c^3(64R^2V^2)\},$$

so that

$$Q_{10} = -c^6R^2(T^2 - 64S^3),$$

and therefore

$$64S^3 - T^2 = \frac{Q_{10}}{c^6R^2},$$

formulae which are interesting in the theory.

As regards the discriminant Q_{10} , this as already remarked, has been calculated for the general form, but for the present purpose it is easier, by dealing directly with the form $[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2 y^2$ and then interpreting D, V, Θ, R and restoring the coefficient c as above, to obtain the discriminant Q_{10} of the function $[c(a, h, k, bx, y)^2]^2 + 4[(j, l, fx, y)^2]^2$ in the required form, as a function of $c^2 D, V, c\Theta, cR$.

I find after some laborious calculations

$$\begin{aligned}
Q_5 &= \text{No. 31} = c^5\{90\} + c^4\{40VD - 288V\Theta\} + c^3\{256V^2\}, \\
Q_7 &= 10000 \text{ No. 34} = c^7\{-99D^2 - 400RD + 2304V\Theta D + 8640\Theta^2\} \\
&\quad + c^6\{12800R^2V\Theta + 82944V^3\Theta^2 - 4608V^2D\} + c^5\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\}, \\
Q_8 &= 1000000 \text{ No. 35} = c^8\{7992D^3 + 72000RD^2 - 145152V\Theta D^2 + 622080\Theta^2D\} \\
&\quad + c^7\{160000R^2\Theta + 691200R^2V\Theta D + 3456000R^2\Theta^2 + 3815424V^2\Theta^2D \\
&\quad + 36080640V\Theta^3 + 635904V^2D^2\} \\
&\quad + c^6\{33177600R^2V^2\Theta^2 + 4669440R^2V^2D + 217645056V^4\Theta^2 + 23003136V^2\Theta D\} \\
&\quad + c^5\{62914560R^2V^4 + 452984832V^3\Theta + 14155776V^4D\} \\
&\quad + c^4\{134217728V^8\}.
\end{aligned}$$

Also $Q_{10} = \text{multiple of discriminant} = c^{10}R^2U^2$, where

$$\begin{aligned}
U &= c^4D^3 + c^3\{-8R^2D - 24R^2V\Theta D + 64R^2\Theta^3\} \\
&\quad + c^2\{-16R^2 - 96R^2V\Theta + 48R^2V^2\Theta^2 - 16R^2V^3D\} + c\{-64R^2V^2\}.
\end{aligned}$$

To these may be joined the intermediate values:

$$\begin{aligned}
Q_3 &= c^3\{81D\} + c^2\{-720RD + 5184V\Theta D + 1600R^2 + 23040RV\Theta + 82944V^2\Theta^2 + 4608V^2D\} \\
&\quad + c\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\}, \\
Q_4 &= c^4\{180D^2 + 1120RD + 2880V\Theta D - 8640\Theta^2\} \\
&\quad + c^3\{1600R^2 + 10240RV\Theta + 270D^2 + 720RD^2 - 51840V\Theta D^2 - 77760\Theta^2D\}.
\end{aligned}$$

We have

$$c(fO) = T - 12VS + 4V^3 - 4cR,$$

and if by means of these values we eliminate $c\Theta$ and c^2D , we obtain Q_5 , Q_7 , Q_8 and Q_{10} as functions of S , T , V and cR . Choosing instead of Q_5 and Q_8 the combinations $Q_5 - Q_3$ and $Q_8 - 8Q_4^2$, and forming also the expression for the combination $Q_5(Q_5 - Q_3)$, we have thus the system of formulae:

$$\begin{aligned} Q_3 &= 9T^2 + 180VS + 4V^3 + 4cR, \\ \frac{1}{4}(Q_5 - Q_3) &= 9T^2 - 432S^3 - 72TVS - 72TV - 16TcR + 864V^2S^2 + 144V^2S + 128VScR, \\ \frac{1}{4}(Q_8 - 8Q_4^2) &= 27T^4 - 4212T^2V^2S^2 + 6588T^2V^2 + 252T^2cR - 7776TS^3 \\ &\quad - 16848TV^2S^2 + 3456TV^2S - 2592TV^2ScR - 1296TV - 1824TV^2cR \\ &\quad - 448T^2c^2R^2 + 544320VS^4 - 461376V^2S^3 + 74304cRS^3 + 15552V^2S^2 \\ &\quad - 10368V^2S^2cR - 10728V^2S - 3264V^2ScR - 1536V^2c^2R^2, \\ &\quad + 81T^4 + 972T^3VS - 612T^3V - 108T^3cR - 3888TS^3 - 5184TV^2S^2 \\ &\quad - 11952TV^2S - 2016TVScR - 288TV - 352TV^2cR - 64Tc^2R \\ &\quad - 77760VS^4 + 153792V^2S^3 - 1728cRS^3 + 29376V^2S^2 + 26928V^2S^2cR \\ &\quad + 576V^2S + 1088V^2ScR + 512V^2c^2R^2. \end{aligned}$$

And, as mentioned above, $Q_{10} = -c^6R^2(T^2 - 64S^3)$.

The just-mentioned value of Q_{10} should, I think, admit of being established *a priori*, and if this be so, then the substitution of the values of S and T in terms of c^2D , V , $c\Theta$, cR , would be the easiest way of arriving at the before-mentioned expression of Q_{10} in terms of these same quantities. The calculation by which this expression was arrived at, is however not without interest, and it will be as well to indicate the mode in which it was effected.

Calculation of Q_{10} .

We have to find the discriminant of $c[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2 y^2$.

Consider for a moment the more general form $P^2 + 4Q^2$, then to find the discriminant, we have to eliminate between the equations

$$P \frac{\partial P}{\partial x} + 4Q \frac{\partial Q}{\partial x} = 0, \quad P \frac{\partial P}{\partial y} + 4Q \frac{\partial Q}{\partial y} = 0;$$

these are satisfied by the system $P = 0, Q^2 = 0$, and it follows that if R be the resultant of the equations $P = 0, Q = 0$, then the discriminant in question contains the factor R^2 . For the other factor we may reduce the system to

$$P \frac{\partial P}{\partial x} + 4Q \frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial x} \frac{\partial Q}{\partial y} - \frac{\partial P}{\partial y} \frac{\partial Q}{\partial x} = 0.$$

Now writing $Q = 2lxy$, these equations become

$$P \frac{\partial P}{\partial x} + 4 \cdot 8l^2 xy^2 = 0, \quad \left(\frac{\partial P}{\partial x} \frac{\partial P}{\partial y} \right) = 0,$$

the resultant of which is = resultant of the system

$$P^2 + 4 \cdot 8l^2 x^2 y^2 = 0, \quad \frac{\partial P}{\partial x} \frac{\partial P}{\partial y} = 0,$$

but in virtue of the second equation, we have

$$\frac{\partial P}{\partial x} \frac{\partial P}{\partial y} = 0,$$

which reduces the first equation to

$$P^2 + 32l^2x^2y^2 = 0,$$

or omitting the factor $2y$, to

$$\frac{\partial P}{\partial x} = 0.$$

Hence, writing $P = \sqrt{c} \cdot (a, h, k, bx, y)^2$, and therefore $\frac{\partial P}{\partial x} = 3\sqrt{c} \cdot (a, h, k, bx, y)^2$, $\frac{\partial P}{\partial y} = 3\sqrt{c} \cdot (h, k, bx, y)^2$; writing also $y = 1$, the two equations become

$$\begin{aligned} c(a, h, k, bx, 1)^2(h, k, bx, 1)^2 + 8l^2x^2 &= 0, \\ (a, h, k, bx, 1)^2 - (h, k, bx, 1)^2 &= 0, \end{aligned}$$

the second of which is more simply written

$$(a, h, -k, -bx, 1)^2 = 0.$$

Hence, restoring the factor P , and also to avoid fractions introducing the factor $8a^2$, the resultant of the two equations is

$$= 8l^2a^2 \prod (8l^2x^2 + c(a, h, k, bx, 1)^2(h, k, bx, 1)^2),$$

where $\prod$ denotes the product of the factors corresponding to the three roots x_1, x_2, x_3 of the equation

$$(a, h, -k, -bx, 1)^2 = 0, \quad \text{or} \quad ax^3 + hx^2 - kx - b = 0,$$

so that the symmetric functions are to be found from

$$\Sigma x_1 = -\frac{h}{a}, \quad \Sigma x_1x_2 = -\frac{k}{a}, \quad x_1x_2x_3 = \frac{b}{a}.$$

The required discriminant is the foregoing resultant multiplied by R , or say by c^2E^2 , that is the discriminant Q_{10} is

$$= c^2E^2 \cdot 8l^2a^2 \prod (8l^2x^2 + \Omega),$$

if for shortness we write

$$\Omega = (a, h, k, bx, 1)^2 \cdot (a, k, bx, 1)^2,$$

and when the symmetric functions have been expressed in terms of the coefficients, the result is to be expressed as a function of D, V, Θ, R by means of the values

$$\begin{aligned} c^2D &= a^2b^2 + 4ahk^2 + 4bk^2h - 6abhk - 3h^2k^2, \\ c\Theta &= -l(ab - hk), \\ cR &= -81l^2ab. \end{aligned}$$

Thus, for instance, the first term of the result is

$$= c^2R^2 \cdot 8l^2a^2 \cdot 512l^6x_1^2x_2^2x_3^2,$$

which is

$$= c^2R^2 \cdot 4096l^8 \cdot \frac{b^2}{a^2} = -64c^4R^4V^2,$$

which is a term in the before-mentioned expression for Q_{10} .

401. A Notation of the Points and Lines in Pascal's Theorem

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 268—274.]

Taking six points 1, 2, 3, 4, 5, 6 on a conic; let $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O$ denote each a combination of three lines, thus

$$\begin{array}{lll}
 12 \cdot 34 \cdot 56 = A & 13 \cdot 45 \cdot 62 = B & 14 \cdot 56 \cdot 23 = C \\
 15 \cdot 62 \cdot 34 = D & 16 \cdot 23 \cdot 45 = E & 12 \cdot 35 \cdot 64 = F \\
 13 \cdot 46 \cdot 25 = G & 14 \cdot 52 \cdot 36 = H & 15 \cdot 63 \cdot 42 = I \\
 16 \cdot 24 \cdot 53 = J & 12 \cdot 36 \cdot 45 = K & 13 \cdot 42 \cdot 56 = L \\
 14 \cdot 53 \cdot 62 = M & 15 \cdot 64 \cdot 23 = N & 16 \cdot 25 \cdot 34 = O
 \end{array}$$

then any hexagon formed with the six points may be represented by a combination of some two of the letters A, B , &c., viz. the three alternate sides are the lines represented by one letter, and the other three alternate sides the lines represented by the other letter: for example, the hexagon 123456 is AE ; and so for the other hexagons.

Any duad AE thus representing a hexagon may be termed a *hexagonal duad*; the number of such duads is sixty.

Each Pascalian line may be denoted by the symbol of the hexagon to which it belongs; thus, the line which belongs to the hexagon AE , is the line AE .

I form the following combinations:

$IMO \cdot DHJ$	each involving all the duads 12, &c. except those of 123 · 456,
$DEG \cdot BNO$	124 · 356,
$ELM \cdot BGJ$	125 · 346,
$HLN \cdot CGI$	126 · 345,
$EFI \cdot JKN$	134 · 256,
$AEH \cdot GKO$	135 · 246,
$AMN \cdot GDF$	136 · 245,
$AGJ \cdot ELO$	145 · 236,
$ABI \cdot DKL$	146 · 235,
$OKM \cdot BFH$	156 · 234,

and also the combinations:

$AEGMI$	involving all the duads 12, 13, &c.,
$ABHJN$	
$BGFIO$	
$CDGJK$	
$DEFHL$	
$KLMNO$	

which I call respectively the *ten-partite* and *six-partite* arrangements.

It is to be remarked that (considering $IMO \cdot DHJ$ as standing for the six duads IM, IO, MO, DH, DJ, HJ , and so for the others) the ten-partite arrangement contains all the sixty hexagonal duads: and in like manner, (considering $AEGMI$ as standing for the ten duads $AE, AG, AM, AI, EG, EM, EI, GM, GI, MI$, and so for the others) the six-partite arrangement contains all the sixty hexagonal duads.

The 60 Pascalian lines intersect by 4's in the 45 Pascalian points p , by 3's in 20 points g and in 60 points h , and by 2's in 90 points m , 360 points r , 360 points t , 360 points z , and 90 points w .

The intersections of the Pascalian lines thus are

$45p$	counting as	270
$20g$	"	40
$60h$	"	180
$90m$	"	90
$360r$	"	360
$360t$	"	360
$360z$	"	360
$90w$	"	90
$1770 = 60 \cdot 59,$		

and the intersections on each Pascalian line are

$9p$	counting as	9
$2g$	"	2
$3h$	"	6
$3m$	"	3
$12r$	"	12
$12t$	"	12
$12z$	"	12
$3w$	"	3
59.		

For the ten-partite arrangement, any double triad such as $ABI \cdot DKL$ gives 15 intersections; $10 \times 15 = 150$; and any pair of double triads such as $ABI \cdot DKL$ and $AEH \cdot GKO$ gives 36 intersections; $45 \times 36 = 1620$; and these are

$$\begin{array}{r}
 10 \times 6g = 60g \quad 10 \times 15p = 150 \\
 45 \times 2h = 90h \quad 45 \times 36 = 1620 \\
 180h \\
 8r = 360r \\
 8t = 360t \\
 8z = 360z \\
 2m = 90w \\
 \hline
 150 \quad 36 \\
 1620 \\
 1770.
 \end{array}$$

For the six-partite arrangement any pentad such as $ABHJN$ gives 45 intersections; $6 \times 45 = 270$; and any two pentads such as $ABHJN$ and $AEGMI$ give 100 intersections; $15 \times 100 = 1500$; and these are

$$\begin{array}{r}
 6 \times 15p = 90 \quad 15 \times 6h = 90h \\
 60g \quad 15 \times 24r = 360r \\
 180h \quad 15 \times 24t = 360t \\
 90m \quad 15 \times 24z = 360z \\
 45 \times 2 = 90w \\
 \hline
 15 \times 18p = 270 \quad 100 \\
 1500 \\
 1770.
 \end{array}$$

I analyse the intersections of a Pascalian line, say AE , by the remaining 59 Pascalian lines as follows:

Observe that AE belongs to the triad AEH , the complementary triad whereof is GKO ; it also belongs to the pentad $AEIMG$. We thus obtain, corresponding to AE , the arrangement

$$\begin{array}{cccccc}
 H & & H & H & H & A \\
 B & & N & J & & \\
 \text{cline1-3 } E & F & L & D & & \\
 \text{cline1-4 } I & M & G & & K & \\
 & & & C & O &
 \end{array}$$

viz. $HABNJ$, is the pentad which contains HA , the arrangement of the last three letters B, N, J thereof being arbitrary; $HEFLD$ is the pentad that contains HE , but the last three letters are so arranged that the columns HBF, HNL, HJD are each of them a triad, IMG is then the residue of the pentad $AEIMG$, and KGO is the complementary triad to AEH , but the arrangement of the letters IMG , and of the letters KGO , are each of them determinate; viz. these are such that we have $BFIGO, NLMKO, JDGCK$, each of them a pentad.

And this being so we derive from the arrangement

$$\begin{array}{ll}
2g & AH, EH; \\
3m & KG, KC, GO; \\
6h & AI, AM, AG; EI, EM, EG; \\
12z & IB, IF, MN, ML, GJ, GD; \\
& HB, HF, HN, HL, HJ, HD; \\
9p & AB, AN, AJ; EF, EL, ED; BF, NL, JD; \\
12r & CB, OF, GJ, GD; OB, OF, ON, OL; \\
& KN, KL, KJ, KD; \\
& FL, FD, LD; BN, BJ, NJ; \\
12t & IG, IO; MK, MO; GK, GC; \\
3m & IM, IC, MG; \\
3w & CD, EK, LM;
\end{array}$$

viz. the line AE in question meets AH, EH each of them in a point g ; KG, KC, GO each in a point m ; and so on.

By constructing in the same way an arrangement for each of the lines AH , &c., we find the nature of the point of intersection of any two of the lines AB, AE, AH , &c.; and we may then present the results in a table (see Plate), which shows at a glance what is the point of intersection (whether a point g, m, h, z, p, r, t , or w) of any two of the Pascalian lines.

I further remark that representing the 45 Pascalian points as follows:

$$\begin{array}{llllll}
12 \cdot 34 = a & 12 \cdot 35 = b & 12 \cdot 36 = c & 12 \cdot 45 = d & 12 \cdot 46 = e & 12 \cdot 56 = f \\
23 \cdot 45 = c & 23 \cdot 46 = g & 23 \cdot 56 = h & 24 \cdot 35 = j & 24 \cdot 36 = k & 24 \cdot 56 = l \\
13 \cdot 24 = g & 13 \cdot 25 = h & 13 \cdot 26 = i & 13 \cdot 45 = j & 13 \cdot 46 = k & 13 \cdot 56 = l \\
25 \cdot 34 = l & 25 \cdot 36 = m & 25 \cdot 46 = n & 26 \cdot 34 = p & 26 \cdot 35 = q & 26 \cdot 45 = r \\
14 \cdot 23 = m & 14 \cdot 25 = n & 14 \cdot 26 = o & 14 \cdot 35 = p & 14 \cdot 36 = q & 14 \cdot 56 = r \\
34 \cdot 56 = p & 35 \cdot 46 = s & 36 \cdot 45 = t & & & \\
15 \cdot 23 = s & 15 \cdot 24 = t & 15 \cdot 26 = u & 15 \cdot 34 = v & 15 \cdot 36 = w & 15 \cdot 46 = x \\
16 \cdot 23 = y & 16 \cdot 24 = z & 16 \cdot 25 = \alpha & 16 \cdot 34 = \beta & 16 \cdot 35 = \gamma & 16 \cdot 45 = \delta
\end{array}$$

the sixty hexagons and their Pascalian lines then are

<i>AE</i>	123456	12 · 45 23 · 56 34 · 61	$dv\pi$
<i>AH</i>	125634	12 · 63 25 · 34 56 · 41	$c\lambda r$
<i>EH</i>	145236	14 · 23 45 · 36 52 · 61	mqa
<i>CK</i>	123654	12 · 65 23 · 54 36 · 41	M
<i>CO</i>	143256	14 · 25 43 · 56 32 · 61	$np\gamma$
<i>KO</i>	125436	12 · 43 25 · 36 54 · 61	$a\mu h$
<i>AM</i>	126534	12 · 53 26 · 34 65 · 41	$b\lambda r$
<i>AG</i>	125643	12 · 64 25 · 43 56 · 31	$e\lambda l$
<i>AI</i>	124365	12 · 36 24 · 65 43 · 51	ckv
<i>EG</i>	132546	13 · 54 32 · 46 25 · 61	$j^?a$
<i>BF</i>	126435	12 · 43 26 · 35 64 · 51	amx
<i>FL</i>	124653	12 · 65 24 · 53 46 · 31	fOk
<i>DL</i>	134265	13 · 26 34 · 65 42 · 51	ipt
<i>BN</i>	132645	13 · 64 32 · 45 26 · 51	$k\epsilon u$
<i>BJ</i>	135426	13 · 42 35 · 26 54 · 61	$g\omega\xi$
<i>JN</i>	153246	15 · 24 53 · 46 32 · 61	$t\alpha\gamma$
<i>GK</i>	125463	12 · 46 25 · 63 54 · 31	$e/\eta j$
<i>KM</i>	126354	12 · 35 26 · 54 63 · 41	$hurq$
<i>IO</i>	152436	15 · 43 52 · 36 24 · 61	$\nu\epsilon\alpha$
<i>MO</i>	143526	14 · 52 43 · 26 35 · 61	$n\zeta h$
<i>EM</i>	145326	14 · 32 45 · 26 53 · 61	$r\iota\pi'\zeta$
<i>EI</i>	154236	15 · 23 54 · 36 42 · 61	$s\tau\xi$
<i>AN</i>	123465	12 · 46 23 · 65 34 · 51	$e\eta\nu$
<i>AJ</i>	124356	12 · 35 24 · 56 43 · 61	$b\kappa\lambda$
<i>AB</i>	126543	12 · 54 26 · 43 65 · 31	$d\zeta l$
<i>DE</i>	154326	15 · 32 54 · 26 43 · 61	$s\tau\tau'\zeta$
<i>EL</i>	132456	13 · 45 32 · 56 24 · 61	$j\eta\xi$
<i>EF</i>	123546	12 · 54 23 · 46 35 · 61	$d\phi\gamma$
<i>CD</i>	143265	14 · 26 43 · 65 32 · 51	$op\xi$
<i>CF</i>	123564	12 · 56 23 · 64 35 · 41	$A\rho$

<i>CG</i>	132564	13 · 56 32 · 64 25 · 41	$l\mu n$
<i>CI</i>	142365	14 · 36 42 · 65 23 · 51	$q\iota c\xi$
<i>MN</i>	146235	14 · 23 46 · 35 62 · 51	$ma\alpha$
<i>GJ</i>	135246	13 · 24 35 · 46 52 · 61	$ga\alpha L$
<i>BI</i>	136245	13 · 24 36 · 45 62 · 51	gru
<i>DG</i>	134625	13 · 62 34 · 25 46 · 51	$i\kappa\alpha\xi$
<i>LM</i>	135624	13 · 62 35 · 24 56 · 41	$i\lambda r$
<i>FI</i>	124635	12 · 63 24 · 35 46 · 51	cdx
<i>BH</i>	136254	13 · 25 36 · 54 62 · 41	$hr\alpha$
<i>FH</i>	125364	12 · 36 25 · 64 53 · 41	cvp
<i>FO</i>	125346	12 · 34 25 · 46 53 · 61	avy
<i>LO</i>	134256	13 · 25 34 · 56 42 · 61	hpz
<i>DK</i>	126345	12 · 34 26 · 45 63 · 51	$airw$
<i>KL</i>	124563	12 · 56 24 · 63 45 · 31	$f\xi j$
<i>BO</i>	134526	13 · 52 34 · 26 45 · 61	$h\beta\xi$
<i>NO</i>	152346	15 · 34 52 · 46 23 · 61	νvy
<i>BC</i>	132654	13 · 65 32 · 54 26 · 41	leo
<i>CJ</i>	142356	14 · 35 42 · 56 23 · 61	$p\kappa y$
<i>JK</i>	124536	12 · 53 24 · 36 45 · 61	$h\delta$
<i>KN</i>	123645	12 · 64 23 · 45 36 · 51	$e\epsilon w$
<i>DH</i>	143625	14 · 62 43 · 25 36 · 51	$o\lambda w$
<i>HJ</i>	142536	14 · 53 42 · 36 25 · 61	$\rho\iota a$
<i>HL</i>	136524	13 · 52 36 · 24 65 · 41	hir
<i>HN</i>	146325	14 · 32 46 · 25 63 · 51	$m\nu w$
<i>BF</i>	126453	12 · 46 26 · 53 64 · 31	$dc\alpha k$
<i>DJ</i>	153426	15 · 42 53 · 26 34 · 61	$t\alpha\xi$
<i>LN</i>	132465	13 · 46 32 · 65 24 · 51	$k\eta t$
<i>GM</i>	135264	13 · 26 35 · 64 52 · 41	$i\iota r n$
<i>IM</i>	142635	14 · 63 42 · 35 26 · 51	qdv
<i>GI</i>	136425	13 · 42 36 · 25 64 · 51	$g\mu M$

Each Pascalian point belongs to four different hexagons; viz.

<i>a</i> to the hexagons	$\{K, F\}\{D, O\}$	<i>n</i>	$\{B, L\}\{F, N\}$
<i>b</i>	$\{A, K\}\{M, J\}$	<i>o</i>	$\{A, C\}\{B, G\}$
<i>c</i>	$\{A, F\}\{H, I\}$	<i>p</i>	$\{B, D\}\{G, H\}$
<i>d</i>	$\{A, F\}\{B, E\}$	<i>q</i>	$\{C, M\}\{I, K\}$
<i>e</i>	$\{A, K\}\{G, N\}$	<i>r</i>	$\{A, L\}\{H, M\}$
<i>f</i>	$\{G, L\}\{K, F\}$	<i>s</i>	$\{C, E\}\{D, I\}$
<i>g</i>	$\{B, O\}\{I, J\}$	<i>t</i>	$\{J, L\}\{D, N\}$
<i>h</i>	$\{B, L\}\{H, O\}$	<i>u</i>	$\{B, M\}\{I, N\}$
<i>i</i>	$\{D, M\}\{G, L\}$	<i>v</i>	$\{A, O\}\{N, I\}$
<i>j</i>	$\{E, K\}\{C, L\}$	<i>w</i>	$\{D, N\}\{H, K\}$
<i>k</i>	$\{B, L\}\{F, N\}$	<i>x</i>	$\{D, I\}\{F, G\}$
<i>l</i>	$\{A, C\}\{B, G\}$	<i>y</i>	$\{C, N\}\{J, O\}$
<i>m</i>	$\{E, N\}\{H, M\}$	<i>z</i>	$\{E, G\}\{I, L\}$
<i>n</i>	$\{C, M\}\{G, O\}$	α	$\{E, J\}\{G, H\}$
<i>o</i>	$\{A, C\}\{B, G\}$	β	$\{A, D\}\{E, J\}$
<i>p</i>	$\{G, H\}\{F, J\}$	γ	$\{E, O\}\{F, M\}$
<i>q</i>	$\{C, M\}\{I, K\}$	δ	$\{B, K\}\{J, O\}$
<i>r</i>	$\{A, L\}\{H, M\}$	ϵ	$\{B, K\}\{C, N\}$
<i>s</i>	$\{C, E\}\{D, I\}$	ζ	$\{C, E\}\{F, G\}$
<i>t</i>	$\{J, L\}\{D, N\}$	η	$\{A, L\}\{E, N\}$
<i>u</i>	$\{B, M\}\{I, N\}$	θ	$\{F, M\}\{I, L\}$
<i>v</i>	$\{A, O\}\{N, I\}$	ι	$\{H, K\}\{J, L\}$
<i>w</i>	$\{D, N\}\{H, K\}$	κ	$\{A, C\}\{I, J\}$
<i>x</i>	$\{D, I\}\{F, G\}$	λ	$\{A, D\}\{G, H\}$
<i>y</i>	$\{C, N\}\{J, O\}$	μ	$\{G, O\}\{D, K\}$
<i>z</i>	$\{E, G\}\{I, L\}$	ν	$\{F, N\}\{H, O\}$
α	$\{E, J\}\{G, H\}$	ξ	$\{A, O\}\{B, M\}$
β	$\{A, D\}\{E, J\}$	<i>o</i>	$\{B, E\}\{F, J\}$
γ	$\{E, O\}\{F, M\}$	π	$\{A, M\}\{E, K\}$
δ	$\{B, K\}\{J, O\}$	ρ	$\{C, L\}\{D, O\}$
ϵ	$\{B, K\}\{C, N\}$	σ	$\{O, N\}\{J, M\}$
ζ	$\{C, E\}\{F, G\}$	τ	$\{B, E\}\{H, I\}$

I have constructed on a very large scale a figure of the sixty Pascalian lines, and the forty-five Pascalian points, marking them according to the foregoing notation; but the figure is from its complexity, and the inconvenient way in which the points are either crowded together or fly off to a great distance, almost unintelligible.

[The two plates (I and II) show the complete table of intersections of the 60 Pascalian lines. Each entry shows the type of point (*g*, *m*, *h*, *z*, *p*, *r*, *t*, or *w*) at which any two Pascalian lines meet.]

402. On a Singularity of Surfaces

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 332—338.]

A SURFACE having a nodal line has in general on this nodal line points where the two tangent planes coincide, or as I propose to term them “pinch-points.” Thus, if the nodal line be the curve of complete intersection of any two surfaces $P = 0$, $Q = 0$, then the equation of the general surface having this curve for a nodal line is

$$(a, b, cP, Q)^2 = 0$$

(where a, b, c are any functions of the coordinates), and the pinch-points are given as the intersections of the nodal line $P = 0$, $Q = 0$ with the surface $ac - b^2 = 0$.

Consider the case where the nodal curve is a curve of partial intersection represented by the equations

$$\frac{P}{P'} = \frac{Q}{Q'} = \frac{R}{R'}, \quad \text{or say by the equations } p = 0, \quad q = 0, \quad r = 0,$$

(viz. p, q, r denote the functions $QR' - Q'R$, $RP' - R'P$, $PQ' - P'Q$ respectively), and consequently we have identically

$$(P, Q, Rp, q, r)^2 = 0, \quad (P', Q', R'p, q, r)^2 = 0,$$

or what is the same thing, (λ, μ) being arbitrary,

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R'p, q, r)^2 = 0.$$

The general surface having the curve in question for its nodal line is represented by the equation

$$(a, b, c, f, g, hp, q, r)^2 = 0,$$

(where (a, b, c, f, g, h) are any functions of the coordinates), and it is easy to see that the condition for a pinch-point is the same as that which (considering p, q, r as coordinates and all the other quantities as constants), expresses that the line

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R'p, q, r)^2 = 0$$

touches the conic

$$(a, b, c, f, g, hp, q, r)^2 = 0,$$

viz. A, B, C, F, G, H being the inverse coefficients, $A = bc - f^2$, &c., this condition is

$$(A, B, C, F, G, H\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R')^2 = 0,$$

or what is the same thing, the pinch-points are given as the common intersections of the nodal line $p = 0$, $q = 0$, $r = 0$ with each of the three surfaces

$$\begin{aligned} (A, B, C, F, G, HP, Q, R)^2 &= 0, \\ (A, B, C, F, G, HP, Q, R)(P', Q', R') &= 0, \\ (A, B, C, F, G, HP', Q', R')^2 &= 0, \end{aligned}$$

these last three equations in fact, adding only a single relation to the relations expressed by the equations $p = 0$, $q = 0$, $r = 0$.

If the functions P, Q, R, P', Q', R' are linear functions of the coordinates, then the nodal curve is a cubic curve in space, the equations $p = 0$, $q = 0$, $r = 0$ represent each of them a quadric surface, and the curve is the partial intersection of any two of these quadric surfaces. The pinch-points are the intersections of the cubic curve with the three quadric surfaces represented by the last-mentioned equations; and since the cubic curve lies on each of the three quadric surfaces $p = 0$, $q = 0$, $r = 0$, the number of pinch-points is at once seen to be = 12.

Original scan reference pages

The following pages reproduce the corresponding range of the Internet Archive scan for visual reference of figures, tables, and any text that may have been compressed in the typeset reconstruction. Source: `collectedmathema06cayluoft.pdf`, pages 101–150.

It is interesting to verify this; I write $Z = -X - Y$, the cubic function then assumes the form

$$(\alpha, \beta, \gamma, \delta) X, Y)^3,$$

where $(\alpha, \beta, \gamma, \delta)$ have the values presently given.

The Hessian is

$$(2\alpha\gamma - 2\beta^2) X^2 + (\alpha\delta - \beta\gamma) \cdot 2XY + (2\beta\delta - 2\gamma^2) Y^2,$$

or writing $2XY = Z^2 - X^2 - Y^2$, this is

$$= (2\alpha\gamma - 2\beta^2 - \alpha\delta + \beta\gamma) X^2 + (2\beta\delta - 2\gamma^2 - \alpha\delta - \beta\gamma) Y^2 + (\alpha\delta - \beta\gamma) Z^2.$$

We find after some easy reductions,

$$\begin{aligned} \frac{1}{3}\alpha &= K' + F', & &= -3(a+c)(ac+M), \\ \beta &= -H' + 2K' + I' + F', & &= 3a(a+c)(b+c), \\ -\gamma &= -K' + 2H' + J' + G', & &= 3b(a+c)(b+c), \\ -\frac{1}{3}\delta &= H' + J', & &= -3(b+c)(bc+M), \end{aligned}$$

and hence

$$\alpha\delta - \beta\gamma = -81(a+c)(b+c)\{(ac+M)(bc+M) - ab(a+c)(b+c)\},$$

where the expression in $\{ \}$ is

$$\begin{aligned} &= (ab+2ac+bc)(ab+ac+2bc) - ab(ab+ac+bc+c^2), \\ &= c\{ab(3a+3b) + c(b+2a)(a+2b) - ab(a+b+c)\}, \\ &= 2c(a+b)(bc+ca+ab), \end{aligned}$$

and therefore

$$\alpha\delta - \beta\gamma = -162(b+c)(c+a)(a+b)Mc;$$

the other coefficients may be similarly calculated, and omitting the merely numerical factor, we have

$$\text{Hessian} = (b+c)(c+a)(a+b)M(aX^2 + bY^2 + cZ^2),$$

which is right.

I write next

$$\begin{aligned} \tilde{H}U + 3MU - \mathfrak{S}U &= X^2(-a\mathfrak{S}x + I'Y - F'Z) \\ &+ Y^2(-b\mathfrak{S}y + J'Z - G'X) \\ &+ Z^2(-a\mathfrak{S}z + K'X - H'Y), \end{aligned}$$

or writing $x = z - Y$, $y = z + X$, this is

$$\begin{aligned}\tilde{H}U + 3MU - \mathfrak{S}U = & -\mathfrak{S}z(aX^2 + bY^2 + cZ^2) \\ & + X^2\{(I' + a\mathfrak{S})Y - F'Z\} \\ & + Y^2\{J'Z - (G' + b\mathfrak{S})X\} \\ & + Z^2\{K'X - H'Y\},\end{aligned}$$

we may determine $\mathfrak{S}$, so that the cubic function of X , Y , Z contains the factor $aX^2 + bY^2 + cZ^2$; writing $Z = -X - Y$, then

X^3	$(K' + F')$	$)$	Contains the factor	$(a + c) X^2$	Quotient is	$-3(ac + M) X$
$+ X^2 Y$	$(-H' + 2K' + F' + I' + a\mathfrak{S})$	$)$	+	$2c XY$	+	$3(bc + M) Y$
$+ XY^2$	$(K' - 2H' - G' - J' - b\mathfrak{S})$	$)$	+	$(b + c) Y^2$		
$+ Y^3$	$(-J' - H')$	$)$				

We have seen that

$$\begin{aligned}K' + F' &= -3(a + c)(ac + M), \\ J' + H' &= -3(b + c)(bc + M),\end{aligned}$$

whence the quotient is, as above stated,

$$= -3(ac + M)X + 3(bc + M)Y.$$

Comparing the coefficients of X^2Y , we have

$$\begin{aligned}a\mathfrak{S} &= -(-H' + 2K' + F' + I') + 3(a + c)(bc + M) - 6c(ac + M), \\ &= 9a(a + c)(b + c) + 3(a + c)(ab + ac + 2bc) - 6c(ab + 2ac + bc), \\ &= 12a(bc + ca + ab) = 12aM,\end{aligned}$$

that is $\mathfrak{S} = 12M$; and the same value would have been obtained by comparing the coefficients of XY^2 . Hence $\tilde{H}U - \mathfrak{S}MU$ divides by $aX^2 + bY^2 + cZ^2$, the quotient being

$$-12Mz - 3(ac + M)X + 3(bc + M)Y,$$

which is

$$= -12Mz - 3(ac + M)(y - z) + 3(bc + M)(z - x),$$

or, finally it is

$$= -3\{(bc + M)x + (ca + M)y + (ab + M)z\},$$

and we thus have

$$\tilde{H}U - \mathfrak{S}MU = -3(aX^2 + bY^2 + cZ^2) \times \{(bc + M)x + (ca + M)y + (ab + M)z\},$$

so that the three inflexions are the intersections of the cubic curve by the line

$$(bc + M)x + (ca + M)y + (ab + M)z = 0.$$

It may be noticed, that if we write

$$\begin{aligned}x + y + z &= u, \\ bcx + cay + abz &= -Mu, \\ ax + by + cz &= v,\end{aligned}$$

then x, y, z will be as

$$\begin{aligned} & (b-c) \{(2M-bc)u-av\} \\ & : (c-a) \{(2M-ca)u-bv\} \\ & : (a-b) \{(2M-ab)u-cv\}, \end{aligned}$$

and substituting these values in the equation

$$ax(y-z)^2 + by(z-x)^2 + cz(x-y)^2 = 0$$

of the cubic, we have a cubic equation for the ratio $(u:v)$; and thence the values (x, y, z) for the coordinates of the inflexions.

It may be added, that we have

$$\begin{aligned} 12MU = & -3(aX^2 + bY^2 + cZ^2) \{(bc+M)x + (ca+M)y + (ab+M)z\} \\ & + \{X^2(I'Y - F'Z) + Y^2(J'Z - G'X) + Z^2(K'X - H'Y)\} = 0, \end{aligned}$$

which is the equation of the cubic expressed in the canonical form.

Pp. 175—179. Effecting the process indicated p. 73, but writing for greater convenience (x, y, z) in place of (X, Y, Z) , so that the substitution to be made is

$$\begin{aligned} & (a, b, c, f, g, h, i, j, k, l) \\ & = (6x, 6y, 6z, -2y, -2z, -2x, -2z, -2x, -2y, x+y+z), \end{aligned}$$

respectively (where I have corrected a misprint in the formula as originally given) I find the equation of the envelope to be

$$\begin{aligned} & 4yz(y-z)^2 a^4 \\ & + 4zx(z-x)^2 b^4 \\ & + 4xy(x-y)^2 c^4 \\ & + 4zx(z^2 + x^2 + 3yz - 2zx + 5xy) b^3c \\ & + 4xy(x^2 + y^2 + 3zx - 2xy + 5yz) c^3a \\ & + 4yz(y^2 + z^2 + 3xy - 2yz + 5zx) a^3b \\ & + 4xy(x^2 + y^2 + 3yz - 2xy + 5zx) bc^3 \\ & + 4yz(y^2 + z^2 + 3zx - 2yz + 5xy) ca^3 \\ & + 4zx(z^2 + x^2 + 3xy - 2zx + 5yz) ab^3 \\ & + x(x^3 - 2x^2y - 2x^2z + xy^2 + 38xyz + xz^2 + 12y^2z + 12yz^2) b^2c^2 \\ & + y(y^3 - 2y^2z - 2y^2x + yz^2 + 38xyz + yx^2 + 12z^2x + 12zx^2) c^2a^2 \\ & + z(z^3 - 2z^2x - 2z^2y + zx^2 + 38xyz + zy^2 + 12x^2y + 12xy^2) a^2b^2 \\ & + 2yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx) a^2bc \\ & + 2zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy) b^2ca \\ & + 2xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz) c^2ab = 0. \end{aligned}$$

The function on the left-hand side is the quotient by $-48(a+b+c)^3$ of the sextic function, Table 67 of my third Memoir on Quantics, [144]; the foregoing quotient was calculated without using the coefficient of the term in $a^2b^2c^2$ ($\xi\eta^2\zeta^2$) of the table, but by way of verification, I calculated from the table the term in question, and found it to be

$$\begin{aligned} & (x+y+z)^4 \\ & - 2(x+y+z)^2(yz+zx+xy) \\ & + 296(x+y+z)xyz \\ & - 8(y^2z^2+z^2x^2+x^2y^2), \end{aligned}$$

and this should consequently be equal to the coefficient of $a^2b^2c^2$ in the product of $(a+b+c)^3$ into the foregoing quartic function of (a, b, c) that is, it should be

$$\begin{aligned} = & x(x^3-2x^2y-2x^2z+xy^2+38xyz+xz^2+12y^2z+12yz^2) \\ & + y(y^3-2y^2z-2y^2x+yz^2+38xyz+yx^2+12z^2x+12zx^2) \\ & + z(z^3-2z^2x-2z^2y+zx^2+38xyz+zy^2+12x^2y+12xy^2) \\ & + 4yz(11x^2+y^2+z^2-2yz+24xy+24zx) \\ & + 4zx(11y^2+z^2+x^2-2zx+24yz+24xy) \\ & + 4xy(11z^2+x^2+y^2-2xy+24zx+24yz), \end{aligned}$$

which is accordingly found to be the case.

397.

SPECIMEN TABLE $M \equiv a^{\alpha} b^{\beta} \pmod{N}$ FOR ANY PRIME OR COMPOSITE MODULUS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IX. (1868), pp. 95—96 and plate.]

IF N be a prime number, and a one of its primitive roots, then any number M prime to N , or what is the same thing, any number in the series $1, 2, \dots, N-1$, may be exhibited in the form $M \equiv a^{\alpha} \pmod{N}$; where α is said to be the index of M in regard to the particular root a . Jacobi's *Canon Arithmeticus* (Berlin, 1839), contains a series of tables, giving the indices of the numbers $1, 2, 3 \dots N-1$ for every prime number, N less than 1000, and giving conversely for each such prime number the numbers M which correspond to the indices $\alpha = 1, 2, \dots (N-1)$ (*Tabulæ Numerorum ad Indices datos pertinentium et Indicium Numero dato correspondentium*). A similar theory applies, it is well known, to the composite numbers; the only difference is, that in order to exhibit for a given composite number N the different numbers less than N and prime to it, we require not a single root a , but two or more roots $a, b, \dots$ and that in terms of these we have $M \equiv a^{\alpha} b^{\beta} \dots \pmod{N}$. For each root a there is an index A (or say the Indicator of the root), such that $a^A \equiv 1 \pmod{N}$, A being the least index for which this equation is satisfied; and the indices $\alpha, \beta, \dots$ extend from 1 to $A, B, \dots$ respectively; the number of different combinations or the product $AB \dots$, being precisely equal to $\phi(N)$, the number of integers less than N and prime to it. The least common multiple of $A, B, \dots$, is termed the Maximum Indicator, and representing it by I , then for any number M not prime to N , we have $M^I \equiv 1 \pmod{N}$, a theorem made use of by Cauchy for the solution of indeterminate equations of the first order. Thus $N=20$, the roots may be taken to be 3, 11; the corresponding exponents are 4, 2 (viz. $3^4 \equiv 1 \pmod{20}$ $11^2 \equiv 1 \pmod{20}$), and the product of these is 8, the number of integers less than 20 and prime to it; the series [go to p. 86]

[from p. 83] of these is in fact 1, 3, 7, 9, 11, 13, 17, 19, each of which is expressible in the required form, viz. $1 \equiv 3^0 \cdot 11^0$, $3 \equiv 3^1 \cdot 11^0$, $7 \equiv 3^2 \cdot 11^0$, &c. (Mod. 20): the maximum indicator is 4; viz. $1^4 \equiv 1$, $3^4 \equiv 1$, $7^4 \equiv 1$, &c. (Mod. 20).

The table pp. 84, 85 gives the Indices for the numbers less than N and prime to it, for all values of N from 1 to 50; the arrangement may be seen at a glance; of the five lines which form a heading, the first contains the numbers N ; the second the root or roots belonging to each number N , the third the indicators of these roots, the fourth the maximum indicator, the fifth, the number $\phi(N)$. The remaining lines contain the index or indices of each of the ϕN numbers M less than N and prime to it, the number corresponding to such index or indices, being placed outside in the same horizontal line. For example, 30 has the roots 7, 11, indices 4, 2 respectively; the Maximum Indicator is 4, and the number of integers less than 30 and prime to it is 8; taking any such number, say 17, the indices are 1, 1, that is, we have $17 \equiv 7^1 \cdot 11^1 \pmod{30}$.

The foregoing corresponds to the *Tabulæ Indicum Numero dato correspondentium* of Jacobi; on account of multiplicity of roots there does not appear to be any mode of forming a single table corresponding to the *Tabulæ Numerorum ad Indices datos pertinentium*; and there would be no adequate advantage in forming for each number N a separate table in some such form as

$$N = 20.$$

Roots		Nos.
3	11	
0	0	1
0	1	11
1	0	3
1	1	13
2	0	9
2	1	19
3	0	7
3	1	17

which I have written down in the form of a table of single entry; for although (whenever, as in the present case, the number of roots is only two) it might have been better exhibited as a table of double entry, when the number of roots is three or more it could not of course be exhibited as a table of corresponding multiple entry.

398.

ON A CERTAIN SEXTIC DEVELOPABLE, AND SEXTIC SURFACE
CONNECTED THEREWITH.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IX. (1868),
pp. 129—142 and 373—376.]

I PROPOSE to consider [first] the sextic developable derived from a quartic equation, viz. taking this to be $(a, b, c, d, e\tilde{x}t, 1)^4 = 0$, where (a, b, c, d, e) are any linear functions of the coordinates (x, y, z, w) , the equation of the developable in question is

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

I have already, in the paper "On a Special Sextic Developable," *Quarterly Journal of Mathematics*, vol. VII. (1866), pp. 105—113, [373], considered a particular case of this surface, viz. that in which c was $= 0$, the geometrical peculiarity of which is that the cuspidal edge is there an excubo-quartic curve (of a special form, having two stationary tangents), whereas in the general case here considered it is a sextic curve. There was analytically the convenience that the linear functions being only the four functions a, b, d, e , these could be themselves taken as coordinates, whereas in the present case we have the five linear functions a, b, c, d, e .

The developable

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0$$

is a sextic developable having for its cuspidal curve the sextic curve

$$ae - 4bd + 3c^2 = 0,$$

$$ace - ad^2 - b^2e + 2bcd - c^3 = 0,$$

(say $I = 0, J = 0$, as usual), and having besides a nodal curve the equations of which may be written

$$\begin{array}{ccccccc} 6(ac - b^2) & : & 3(ad - bc) & : & ae + 2bd - 3c^2 & : & 3(be - cd) & : & 6(ce - d^2) & : & 9J \\ = & a & : & b & : & c & : & d & : & e & : & I, \end{array}$$

viz. these equations are really equivalent to two equations, and they represent a curve of the fourth order which is an excubo-quartic. We may in fact find the equations of the nodal curve by assuming $(a, b, c, d, e\chi t, 1)^4$ to be a perfect square, say to avoid fractions that it is $= 3(at^2 + 2\beta t + \gamma)^2$, then we have

$$a : b : c : d : e = 3\alpha^2 : 3\alpha\beta : \alpha\gamma + 2\beta^2 : 3\beta\gamma : 3\gamma^2,$$

which equations as involving the two arbitrary parameters $\alpha : \beta : \gamma$, give two equations between (a, b, c, d, e) , and we may at once by means of them verify the above-mentioned equations of the nodal curve. It also hereby appears that the nodal curve is as stated an excubo-quartic curve; viz. we have between a, b, c, d, e a single linear relation, that is a quadric relation between α, β, γ , and this equation may be satisfied identically by taking for α, β, γ properly determined quadric functions of a variable parameter θ ; whence a, b, c, d, e are proportional to quartic functions of the variable parameter θ , or the curve is an excubo-quartic.

The equations of the nodal curve may be presented under a somewhat different form; viz. the cubi-covariant of $(a, b, c, d, e\chi t, 1)^4 = 0$ being

$$\left\{ \begin{array}{l} -a^2d + 3abc - 2b^3 \\ -a^2e - 2abd + 9ac^2 - 6b^2c \\ -5abe + 15acd - 10b^2d \\ + 10ad^2 - 10b^2e \\ + 5ade + 10bd^2 - 15bce \\ + ae^2 + 2bde - 9c^2e + 6cd^2 \\ + be^2 - 3cde + 2d^3 \end{array} \right\} (t, 1)^6 = 0,$$

say this function, multiplied by 6 to avoid fractions, is

$$(a, b, c, d, e, f, g\chi t, 1)^6,$$

that is

$$a = 6(-a^2d + 3abc - 2b^3),$$

$$b = 1(-a^2e - 2abd + 9ac^2 - 6b^2c),$$

$$c = 2(-abe + 3acd - 2b^2d),$$

$$d = 3(+ad^2 - b^2e),$$

$$e = 2(+ade + 2bd^2 - 3bce),$$

$$f = 1(+ae^2 + 2bde - 9c^2e + 6cd^2),$$

$$g = 6(+be^2 - 3cde + 2d^3),$$

then the equations of the nodal curve may be written

$$a = 0, b = 0, c = 0, d = 0, e = 0, f = 0, g = 0.$$

It may be mentioned that we have identically

$$ae - 4bd + 3c^2 = 0,$$

$$af - 3be + 2cd = 0,$$

$$ag - 9ce + 8d^2 = 0,$$

$$bg - 3cf + 2de = 0,$$

$$\cancel{bf - 4ce + 3d^2 = 0}, \quad cg - 4df + 3e^2 = 0.$$

and moreover

$$\begin{aligned} & ag - 6bf + 15ce - 10d^2 \\ &= -6(bf - 4ce + 3d^2) = +6(I^3 - 27J^2), \end{aligned}$$

so that the equation of the developable may be written in the form

$$ag - 6bf + 15ce - 10d^2 = 0,$$

or in the more simple form

$$bf - 4ce + 3d^2 = 0,$$

each of which puts in evidence the nodal curve on the surface.

The nodal and cuspidal curves meet in the points

$$\left\{ \begin{array}{l} \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e}, \end{array} \right.$$

being, as it is easy to show, a system of four points. The four points in question form a tetrahedron, the equations of the faces of which may be taken to be $x=0$, $y=0$, $z=0$, $w=0$; and the equation of the surface may be expressed in this system of quadriplanar coordinates.

We introduce these coordinates *ab initio*, by taking the quartic function of t to be

$$(a, b, c, d, e\chi t, 1)^4 = x(t+\alpha)^4 + y(t+\beta)^4 + z(t+\gamma)^4 + w(t+\delta)^4,$$

that is, by writing

$$a = x + y + z + w,$$

$$b = \alpha x + \beta y + \gamma z + \delta w,$$

$$c = \alpha^2 x + \beta^2 y + \gamma^2 z + \delta^2 w,$$

$$d = \alpha^3 x + \beta^3 y + \gamma^3 z + \delta^3 w,$$

$$e = \alpha^4 x + \beta^4 y + \gamma^4 z + \delta^4 w.$$

Observe that (t_1, t_2, t_3, t_4) being any constant quantities, we thence have

$$\begin{aligned} & e - d\Sigma t_1 + c\Sigma t_1 t_2 - b\Sigma t_1 t_2 t_3 + at_1 t_2 t_3 t_4 \\ &= x(\alpha - t_1)(\alpha - t_2)(\alpha - t_3)(\alpha - t_4) \\ &+ y(\beta - t_1)(\beta - t_2)(\beta - t_3)(\beta - t_4) \\ &+ z(\gamma - t_1)(\gamma - t_2)(\gamma - t_3)(\gamma - t_4) \\ &+ w(\delta - t_1)(\delta - t_2)(\delta - t_3)(\delta - t_4), \end{aligned}$$

and thence in particular

$$e - d\Sigma\alpha + c\Sigma\alpha\beta - b\Sigma\alpha\beta\gamma + a\alpha\beta\gamma\delta = 0,$$

viz. this is the linear relation which subsists identically between (a, b, c, d, e) , the five linear functions of the coordinates (x, y, z, w) .

Starting from the above values of (a, b, c, d, e) , we find without difficulty

$$\begin{aligned} I &= (\alpha - \beta)^4 xy + (\alpha - \gamma)^4 xz + (\alpha - \delta)^4 xw + (\beta - \gamma)^4 yz + (\beta - \delta)^4 yw + (\gamma - \delta)^4 zw, \\ J &= (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2 xyz + (\alpha - \beta)^2 (\beta - \delta)^2 (\delta - \alpha)^2 xyw \\ &\quad + (\alpha - \gamma)^2 (\gamma - \delta)^2 (\delta - \alpha)^2 xzw + (\beta - \gamma)^2 (\gamma - \delta)^2 (\delta - \beta)^2 yzw, \end{aligned}$$

but we thus see the convenience of introducing constant multipliers into the expressions of the four coordinates respectively, viz. writing

$$\begin{aligned} x &= (\beta\gamma\delta)^2 x', \\ y &= (\gamma\delta\alpha)^2 y', \\ z &= (\delta\alpha\beta)^2 z', \\ w &= (\alpha\beta\gamma)^2 w', \end{aligned}$$

where for shortness

$$(\beta\gamma\delta) = (\beta - \gamma)(\gamma - \delta)(\delta - \beta), \text{ \&c.,}$$

or what is the same thing, taking the quartic to be

$$(a, b, c, d, e\sqrt{t}, 1)^4 = x'(\beta\gamma\delta)^2(t + \alpha)^4 + y'(\gamma\delta\alpha)^2(t + \beta)^4 + z'(\delta\alpha\beta)^2(t + \gamma)^4 + w'(\alpha\beta\gamma)^2(t + \delta)^4,$$

we find

$$\begin{aligned} J &= (\lambda'\mu'\nu')^2(x'y'z' + y'z'w' + z'x'w' + x'y'w'), \\ I &= (\lambda'\mu'\nu')\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}, \end{aligned}$$

where for shortness

$$\begin{aligned} \lambda' &= (\alpha - \delta)^2(\beta - \gamma)^2, \\ \mu' &= (\beta - \delta)^2(\gamma - \alpha)^2, \\ \nu' &= (\gamma - \delta)^2(\alpha - \beta)^2, \end{aligned}$$

or writing

$$\begin{aligned} \sqrt{(\lambda')} &= (\alpha - \delta)(\beta - \gamma), \\ \sqrt{(\mu')} &= (\beta - \delta)(\gamma - \alpha), \\ \sqrt{(\nu')} &= (\gamma - \delta)(\alpha - \beta), \end{aligned}$$

we have

$$\sqrt{(\lambda')} + \sqrt{(\mu')} + \sqrt{(\nu')} = 0,$$

and the equation of the developable is thus

$$\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(x'w' + x'y')\}^3 - 27\lambda'\mu'\nu'(x'y'z' + y'z'w' + z'x'w' + x'y'w')^2 = 0.$$

Observe that $J=0$ is a cubic surface passing through each edge of the tetrahedron, and having at each summit a conical point; $I=0$ is a quadric surface passing through each summit of the tetrahedron, and at each of these points the tangent

plane of the quadric surface touches the tangent cone of the cubic surface: to show this it is only necessary to observe that at the point $(x'=0, y'=0, z'=0)$ the tangent cone is $y'z' + z'x' + x'y' = 0$, and the tangent plane is $\lambda'x' + \mu'y' + \nu'z' = 0$, and that these touch in virtue of the above-mentioned relation $\sqrt{(\lambda')} + \sqrt{(\mu')} + \sqrt{(\nu')} = 0$. It follows that on the curve of intersection, or cuspidal edge of the developable, each of the summits is a cuspidal or stationary point, that is, the cuspidal curve has four stationary points; this agrees with the character of the curve as given, Salmon "On the Classification of Curves of Double Curvature," *Camb. and Dubl. Math. Jour.* vol. v. (1850), p. 39, viz. the character is there given

$$a = 6, m = 6, n = 4, r = 6, g = 3, h = 6, \alpha = 0, \beta = 4, x = 4, y = 6,$$

($\beta = 4$, that is, there are as stated 4 stationary points).

To find the equations of the nodal curve, instead of transforming the equations as given in terms of (a, b, c, d, e) , it is better to deduce these from the equation of the surface; viz. if there is a nodal curve, we must have

$$\begin{aligned} \delta_x I : \delta_y I : \delta_z I : \delta_w I : 18J \\ = \delta_x J : \delta_y J : \delta_z J : \delta_w J : I. \end{aligned}$$

Writing these under the form $\delta_x I + \theta' \delta_x J = 0$, &c., where θ' is regarded as an arbitrary parameter⁽¹⁾, we have

$$\begin{aligned} \lambda'w' + \mu'z' + \nu'y' + \theta'(y'z' + y'w' + z'w') &= 0, \\ \lambda'z' + \mu'w' + \nu'x' + \theta'(z'x' + z'w' + x'w') &= 0, \\ \lambda'y' + \mu'x' + \nu'w' + \theta'(x'y' + x'w' + y'w') &= 0, \\ \lambda'x' + \mu'y' + \nu'z' + \theta'(y'z' + z'x' + x'y') &= 0, \end{aligned}$$

which equations (eliminating θ') must be equivalent to two equations only.

I remark that the first three equations may be regarded as a set of linear equations in $1, w', \theta', \theta'w'$; and determining from them the ratios of these quantities, we have, suppose,

$$1 : -w' : \theta' : -\theta'w' = A : B : C : D,$$

where

$$\begin{aligned} A &= \begin{vmatrix} \lambda' & y'z' & y' + z' \\ \mu' & z'x' & z' + x' \\ \nu' & x'y' & x' + y' \end{vmatrix}, & B &= \begin{vmatrix} y'z' & y' + z' & \mu'z' + \nu'y' \\ z'x' & z' + x' & \nu'x' + \lambda'z' \\ x'y' & x' + y' & \lambda'y' + \mu'x' \end{vmatrix}, \\ C &= \begin{vmatrix} y' + z' & \mu'z' + \nu'y' & \lambda' \\ z' + x' & \nu'x' + \lambda'z' & \mu' \\ x' + y' & \lambda'y' + \mu'x' & \nu' \end{vmatrix}, & D &= \begin{vmatrix} \mu'z' + \nu'y' & \lambda' & y'z' \\ \nu'x' + \lambda'z' & \mu' & z'x' \\ \lambda'y' + \mu'x' & \nu' & x'y' \end{vmatrix}. \end{aligned}$$

¹ The value of θ' is in fact $= -\frac{18J}{I}$, that is, instead of the four equations involving an arbitrary parameter θ' , we have really four determinate equations.

We have thence $AD - BC = 0$ and (substituting in the fourth equation) $A(\lambda'x' + \mu'y' + \nu'z') + C(y'z' + z'x' + x'y') = 0$; each of these equations must contain the equation of the cone having $(x' = 0, y' = 0, z' = 0)$ for its vertex, and passing through the nodal curve. The two equations are of the orders 6 and 4 respectively; and as the curve is a quartic curve passing through the vertex in question, the equation of the cone is of the order 3. I have not effected the reduction of the sextic equation, but for the quartic equation, substituting for A, C their values, this is

$$\begin{aligned} & -(\lambda'x' + \mu'y' + \nu'z') [\lambda'x'^2(y' - z') + \mu'y'^2(z' - x') + \nu'z'^2(x' - y')] \\ & + (y'z' + z'x' + x'y') [\lambda'^2x'(y' - z') + \mu'^2y'(z' - x') + \nu'^2z'(x' - y')] \\ & + \{\mu'\nu'(y' - z') + \nu'\lambda'(z' - x') + \lambda'\mu'(x' - y')\} (x' + y' + z') = 0, \end{aligned}$$

which is easily reduced to

$$\begin{aligned} & \lambda'^2x'(-x'^2 + y'z' + z'x' + x'y')(y' - z') \\ & + \mu'^2y'(-y'^2 + y'z' + z'x' + x'y')(z' - x') \\ & + \nu'^2z'(-z'^2 + y'z' + z'x' + x'y')(x' - y') \\ & + \mu'\nu'[(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'](y' - z') \\ & + \nu'\lambda'[(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'](z' - x') \\ & + \lambda'\mu'[(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'](x' - y') = 0; \end{aligned}$$

and I have found that this is transformable into

$$\begin{aligned} & 2\{x'\sqrt{\lambda'} + y'\sqrt{\mu'} + z'\sqrt{\nu'}\} \times [y'z'\sqrt{\lambda'}(\mu'y' - \nu'z')z'x'\sqrt{\mu'}(\nu'z' - \lambda'x')x'y'\sqrt{\nu'}(\lambda'x' - \mu'y')] \\ & - x'y'z'\{\sqrt{\mu'} - \sqrt{\nu'}\}\{\sqrt{\nu'} - \sqrt{\lambda'}\}\{\sqrt{\lambda'} - \sqrt{\mu'}\}] = 0, \end{aligned}$$

viz. the two functions are equivalent in virtue of the relation $\sqrt{\lambda'} + \sqrt{\mu'} + \sqrt{\nu'} = 0$, or, what is the same thing, they only differ by a function $(x', y', z')^4$ into the evanescent factor $\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu'$. The function in $\{ \}$ equated to zero is therefore the equation of the cubic cone.

I do not stop to give the steps of the investigation in the above form, as the investigation may be very much simplified as follows: by linear combinations of the four equations in x', y', z', w', θ' , we deduce

$$\begin{aligned} & (\lambda' - \mu' - \nu')(x' + w' - y' - z') + 2\theta'(y'z' - x'w') = 0, \\ & (-\lambda' + \mu' - \nu')(y' + w' - z' - x') + 2\theta'(z'x' - y'w') = 0, \\ & (-\lambda' - \mu' + \nu')(z' + w' - x' - y') + 2\theta'(x'y' - z'w') = 0, \\ & (\lambda' + \mu' + \nu')(x' + y' + z' + w') + 2\theta'(y'z' + z'x' + x'y' + x'w' + y'w' + z'w') = 0. \end{aligned}$$

Hence writing

$$\begin{aligned} \lambda &= \lambda' - \mu' - \nu', & x &= w' + x' - y' - z', \\ \mu &= -\lambda' + \mu' - \nu', & y &= w' - x' + y' - z', \\ \nu &= -\lambda' - \mu' + \nu', & z &= w' - x' - y' + z', \\ & & w &= w' + x' + y' + z', \end{aligned}$$

we find

$$\mu\nu = \lambda'^2 - \mu'^2 - \nu'^2 + 2\mu'\nu',$$

$$\nu\lambda = -\lambda'^2 + \mu'^2 - \nu'^2 + 2\nu'\lambda',$$

$$\lambda\mu = -\lambda'^2 - \mu'^2 + \nu'^2 + 2\lambda'\mu',$$

and thence

$$\mu\nu + \nu\lambda + \lambda\mu = -(\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu') = 0,$$

that is

$$\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

the relation which connects the new constants λ, μ, ν . Moreover

$$yz - xw = 4(y'z' - x'w'),$$

$$zx - yw = 4(z'x' - y'w'),$$

$$xy - zw = 4(x'y' - z'w'),$$

$$3w^2 - x^2 - y^2 - z^2 = 8(y'z' + z'x' + x'y' + x'w' + y'w' + z'w'),$$

and writing for greater convenience $\theta = -\frac{2}{\theta'}$, the equations are transformed into

$$\theta\lambda x = xw - yz,$$

$$\theta\mu y = yw - zx,$$

$$\theta\nu z = zw - xy,$$

$$2\theta(\lambda + \mu + \nu)w = 3w^2 - x^2 - y^2 - z^2,$$

where

$$\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

viz. these equations, eliminating θ , give the equations of the nodal curve.

From the first three equations eliminating θ , we deduce

$$yzw(\mu - \nu) = x(\mu y^2 - \nu z^2),$$

$$zxw(\nu - \lambda) = y(\nu z^2 - \lambda x^2),$$

$$xyw(\lambda - \mu) = z(\lambda x^2 - \mu y^2),$$

or, as these equations may be written,

$$wxyz = \frac{x^2(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y^2(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z^2(\lambda x^2 - \mu y^2)}{\lambda - \mu},$$

which equations, from the mode in which they are obtained, are it is clear equivalent to two equations only. Using the fourth equation, and eliminating θ by substituting therein for $\theta\lambda, \theta\mu, \theta\nu$ their values from the first three equations, we find

$$2w^2 \left(3w - \frac{yz}{x} - \frac{zx}{y} - \frac{xy}{z} \right) = 3w^2 - x^2 - y^2 - z^2,$$

that is

$$3w^2 + x^2 + y^2 + z^2 = 2w \left(\frac{yz}{x} + \frac{zx}{y} + \frac{xy}{z} \right),$$

or, what is the same thing,

$$xyz(3w^2 + x^2 + y^2 + z^2) - 2w(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

we have to show that this is in fact included in the former system, for then the four equations with θ eliminated will give two equations only.

Observe that the former system may be written

$$w = \frac{x(\mu y^2 - \nu z^2)}{(\mu - \nu)yz},$$

$$(\mu - \nu)y^2z^2 + (\nu - \lambda)z^2x^2 + (\lambda - \mu)x^2y^2 = 0,$$

and that we have thus to show that substituting for w the value $w = \frac{x(\mu y^2 - \nu z^2)}{\mu - \nu}$ in the equation

$$xyz(3w^2 + x^2 + y^2 + z^2) - 2w(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

the result is

$$(\mu - \nu)y^2z^2 + (\nu - \lambda)z^2x^2 + (\lambda - \mu)x^2y^2 = 0.$$

The substitution in question gives

$$\frac{3x^2(\mu y^2 - \nu z^2)^2}{(\mu - \nu)^2 yz} + yz(x^2 + y^2 + z^2) - \frac{2(\mu y^2 - \nu z^2)}{(\mu - \nu)yz}(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

that is

$$3x^2(\mu y^2 - \nu z^2)^2 + (\mu - \nu)^2 y^2 z^2 (x^2 + y^2 + z^2) - 2(\mu - \nu)(\mu y^2 - \nu z^2)(y^2 z^2 + z^2 x^2 + x^2 y^2) = 0,$$

which is in fact

$$-\mu^2 y^2 (y^2 - z^2)(z^2 - x^2) + 2\mu\nu x^2 (y^2 - z^2)^2 - \nu^2 z^2 (y^2 - z^2)(x^2 - y^2) = 0,$$

that is, throwing out the factor $y^2 - z^2$, it is

$$-\mu^2 y^2 (z^2 - x^2) + 2\mu\nu x^2 (y^2 - z^2) - \nu^2 z^2 (x^2 - y^2) = 0.$$

But in virtue of the equation $\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0$, we have

$$\begin{aligned} & \frac{\mu\nu}{\lambda} \{(\mu - \nu)y^2z^2 + (\nu - \lambda)z^2x^2 + (\lambda - \mu)x^2y^2\} \\ &= \frac{\mu\nu}{\lambda} [\lambda x^2(y^2 - z^2) + \mu y^2(z^2 - x^2) + \nu z^2(x^2 - y^2)], \\ &= \mu\nu x^2(y^2 - z^2) - (\mu + \nu)[\mu y^2(z^2 - x^2) + \nu z^2(x^2 - y^2)], \\ &= -\mu^2 y^2(z^2 - x^2) + 2\mu\nu x^2(y^2 - z^2) - \nu^2 z^2(x^2 - y^2), \end{aligned}$$

and the required property thus holds good.

We thus see that the equations of the nodal curve are

$$wxyz = \frac{x^2(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y^3(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z^2(\lambda x^2 - \mu y^2)}{\lambda - \mu},$$

the nodal curve is thus the partial intersection of the two cubic scrolls (skew surfaces)

$$(\mu - \nu)wyz = x(\mu y^2 - \nu z^2), \quad (\nu - \lambda)wzx = y(\nu z^2 - \lambda x^2),$$

viz. taking A, B, C, D to be the summits of the tetrahedron the faces whereof are $x=0, y=0, z=0, w=0$, the first of these has AD for a nodal directrix, BC for a single directrix, BD, CD for generators; the second has BD for a nodal directrix, AC for a single directrix, AD, CD for generators; the surfaces intersect in the line AD twice, the line BD twice, and the line CD ; the order of the residual curve, or nodal curve of the developable, is thus $9 - (2 + 2 + 1) = 4$ as it should be.

I remark that the equation

$$(\mu - \nu)y^2z^2 + (\nu - \lambda)z^2x^2 + (\lambda - \mu)x^2y^2 = 0,$$

is the equation of the cone having its vertex at the point $D, (x=0, y=0, z=0)$, and passing through the nodal curve; the lines DA, DB, DC are each of them a nodal line of the cone, or "line through two points" of the curve; for an excubo-quartic curve the number of lines through two points which pass through a given point not on the curve is in fact = 3.

It remains to introduce the coordinates (x, y, z, w) into the equation of the developable. We have

$$4x' = w + x - y - z,$$

$$4y' = w - x + y - z,$$

$$4z' = w - x - y + z,$$

$$4w' = w + x + y + z,$$

and thence

$$16y'z' = (w - x)^2 - (y - z)^2,$$

$$16x'w' = (w + x)^2 - (y + z)^2,$$

giving

$$8(x'w' + y'z') = w^2 + x^2 - y^2 - z^2$$

and similarly

$$8(y'w' + z'x') = w^2 - x^2 + y^2 - z^2,$$

and

$$8(z'w' + x'y') = w^2 - x^2 - y^2 + z^2.$$

Moreover

$$\begin{aligned} 16(y'z' + z'x' + x'y') &= (w - x)^2 - (y - z)^2 \\ &\quad + (w - y)^2 - (z - x)^2 \\ &\quad + (w - z)^2 - (x - y)^2, \\ &= 3w^2 - 2w(x + y + z) \\ &\quad - x^2 - y^2 - z^2 + 2yz + 2zx + 2xy. \end{aligned}$$

Consequently

$$\begin{aligned}
 64w'(y'z' + z'x' + x'y') &= (w + x + y + z) \\
 &\quad \times \{3w^2 - 2w(x + y + z) - x^2 - y^2 - z^2 + 2yz + 2zx + 2xy\}, \\
 64x'y'z' &= (w + x - y - z)(w - x + y - z)(w - x - y - z), \\
 &= w^3 \\
 &\quad - w^2(x + y + z) \\
 &\quad - w(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) \\
 &\quad + x^3 + y^3 + z^3 - yz^2 - y^2z - zx^2 - z^2x - xy^2 - x^2y + 2xyz.
 \end{aligned}$$

Putting for shortness

$$p = x + y + z, \quad \nabla = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy,$$

the two expressions are

$$\begin{array}{l|l}
 3w^3 & w^3 \\
 + w^2 + p & + w^2 - p \\
 + w - 2p^2 - \nabla & - w \cdot \nabla \\
 - p \nabla & + x^3 + y^3 + z^3 - yz^2 - y^2z - zx^2 - z^2x - x^2y - xy^2 + 2xyz
 \end{array}$$

or observing that $-p\nabla$ is

$$= -x^3 - y^3 - z^3 + y^2z + yz^2 + z^2x + zx^2 + x^2y + xy^2 + 6xyz,$$

we have

$$\begin{aligned}
 64(w'y'z' + w'z'x' + w'x'y' + x'y'z') &= 4w^3 - 2w(p^2 + \nabla) + 8xyz, \\
 &= 4w^3 \\
 &\quad - 4w(x^2 + y^2 + z^2) \\
 &\quad + 8xyz,
 \end{aligned}$$

that is

$$\begin{aligned}
 16(w'y'z' + w'z'x' + w'x'y' + x'y'z') &= w^3 \\
 &\quad - w(x^2 + y^2 + z^2) \\
 &\quad + 2xyz.
 \end{aligned}$$

Moreover

$$\begin{aligned}
 8\{\lambda'(x'w' + y'z') + \mu'(y'x' + z'x') + \nu'(z'w' + x'y')\} &= \lambda'(w^2 + x^2 - y^2 - z^2) \\
 &\quad + \mu'(w^2 - x^2 + y^2 - z^2) \\
 &\quad + \nu'(w^2 - x^2 - y^2 + z^2), \\
 &= -(\lambda + \mu + \nu)w^2 + \lambda x^2 + \mu y^2 + \nu z^2;
 \end{aligned}$$

and we have

$$\lambda = 2(\beta - \delta)(\gamma - \delta)(\gamma - \alpha)(\alpha - \beta),$$

$$\mu = 2(\gamma - \delta)(\alpha - \delta)(\alpha - \beta)(\beta - \gamma),$$

$$\nu = 2(\alpha - \delta)(\beta - \delta)(\beta - \gamma)(\gamma - \alpha),$$

whence $\lambda\mu\nu = 8\lambda'\mu'\nu'$.

Hence finally λ, μ, ν denoting as just mentioned, and therefore satisfying $\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0$, the equation of the developable is

$$\lambda\mu\nu \{w^3 - w(x^2 + y^2 + z^2) + 2xyz\}^2 + 108 \{(\lambda + \mu + \nu)w^2 - \lambda x^2 - \mu y^2 - \nu z^2\}^3 = 0$$

(say this is $\lambda\mu\nu T^2 + 108S^3 = 0$), and this surface (which has obviously the cuspidal curve $S = 0, T = 0$) has also the nodal curve

$$wxyz = \frac{x^2(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y^2(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z^2(\lambda x^2 - \mu y^2)}{\lambda - \mu}.$$

I will show *a posteriori* that this is actually a nodal curve on the surface. Introducing an arbitrary parameter θ , the equations of the curve may be written *ut supra*

$$\theta\lambda x = xw - yz,$$

$$\theta\mu y = yw - zx,$$

$$\theta\nu z = zw - xy,$$

$$2\theta(\lambda + \mu + \nu)w = 3w^2 - x^2 - y^2 - z^2,$$

and we have thence, as before,

$$2w \left(3w - \frac{yz}{x} - \frac{zx}{y} - \frac{xy}{z} \right) = 3w^2 - x^2 - y^2 - z^2.$$

Hence

$$\begin{aligned} 2\theta &= \frac{3w^2 - x^2 - y^2 - z^2}{(\lambda + \mu + \nu)w} = \frac{-2w(x^2 + y^2 + z^2) + 6xyz}{-\lambda x^2 - \mu y^2 - \nu z^2}, \\ &= \frac{3w^2 - w(x^2 + y^2 + z^2)}{(\lambda + \mu + \nu)w^2}, \\ &= \frac{3\{w^3 - w(x^2 + y^2 + z^2) + 2xyz\}}{(\lambda + \mu + \nu)w^2 - \lambda x^2 - \mu y^2 - \nu z^2}, \\ &= \frac{3T}{S}. \end{aligned}$$

Hence writing

$$\theta(-2\lambda x) - (-2wx + 2yz) = 0,$$

$$\theta(-2\mu y) - (-2wy + 2zx) = 0,$$

$$\theta(-2\nu z) - (-2wz + 2xy) = 0,$$

$$\theta \cdot 2(\lambda + \mu + \nu)w - (3w^2 - x^2 - y^2 - z^2) = 0,$$

substituting for θ its value $= \frac{3T}{2S}$, and attending to the significations of S and T , we have

$$3T\delta_x S - 2S\delta_x T = 0,$$

$$3T\delta_y S - 2S\delta_y T = 0,$$

$$3T\delta_z S - 2S\delta_z T = 0,$$

$$3T\delta_w S - 2S\delta_w T = 0,$$

which are in fact the conditions to be satisfied in order that the point (x, y, z, w) may belong to a nodal curve of the surface $\lambda\mu\nu T^2 + 108 S^3 = 0$.

It is to be noticed that the coordinates of the before mentioned four points of intersection of the cuspidal and the nodal curves (being as already mentioned stationary points on the cuspidal curve) may be written $x, y, z, w = (1, 1, 1, 1), (1, -1, -1, 1), (-1, 1, -1, 1), (-1, -1, 1, 1)$.

We have thus far considered the developable, or torse, the equation of which is

$$\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}^3 - 27\lambda'\mu'\nu'(x'y'z' + x'y'w' + x'z'w' + y'z'w')^2 = 0,$$

where $\lambda'(\lambda') + \mu'(\mu') + \nu'(\nu') = 0$; or, what is the same thing, writing a, b, c , in place of $\lambda'(\lambda'), \mu'(\mu'), \nu'(\nu')$ respectively, the torse

$$\{a^2(x'w' + y'z') + b^2(y'w' + z'x') + c^2(z'w' + x'y')\}^3 - 27a^2b^2c^2(x'y'z' + x'y'w' + x'z'w' + y'z'w')^2 = 0,$$

where $a + b + c = 0$.

Inverting this by the equations $x', y', z', w' = \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{w}$, we obtain a sextic surface

$$\{a^2(xw + yz) + b^2(yw + zx) + c^2(zw + xy)\}^3 - 27a^2b^2c^2xyzw(x + y + z + w)^2 = 0,$$

where $a + b + c = 0$; which surface I propose [secondly] to consider in the present paper.

The surface has evidently the singular tangent planes $x = 0, y = 0, z = 0, w = 0$, each osculating the surface in a conic, that is, meeting it in the conic taken thrice, viz.,

$$\begin{array}{llll} x = 0, & \text{in a conic on the quadric cone} & a^2yz + b^2yw + c^2zw = 0, \\ y = 0, & \text{,, ,, ,,} & a^2xw + b^2zx + c^2zw = 0, \\ z = 0, & \text{,, ,, ,,} & a^2xw + b^2yw + c^2xy = 0, \\ w = 0, & \text{,, ,, ,,} & a^2yz + b^2zx + c^2xy = 0; \end{array}$$

and it has also a cuspidal conic, the intersection of the plane $x + y + z + w = 0$ with the quadric surface

$$a^2(xw + yz) + b^2(yw + zx) + c^2(zw + yz) = 0;$$

it may be observed that the four conics of osculation are also sections of this surface.

The surface has also a nodal curve, the equations of which might be obtained by inversion of those of the nodal curve of the sextic torse above referred to; but I prefer to obtain them independently, in a synthetical manner, as follows:

Take α, β, γ arbitrary, and write

$$\begin{aligned} -A &= (b-c)\alpha + b\beta - c\gamma, & F &= b\gamma - c\beta, \\ -B &= (c-a)\beta + c\gamma - a\alpha, & G &= c\alpha - a\gamma, \\ -C &= (b-c)\gamma + a\alpha - b\beta, & H &= a\beta - b\alpha, \\ M &= (b-c)\alpha + (c-a)\beta + (a-b)\gamma, \\ Q &= a^2(b-c)\alpha + b^2(c-a)\beta + c^2(a-b)\gamma; \end{aligned}$$

then it is to be shown, that not only the equation of the surface is satisfied, but that also each of the derived equations is satisfied, by the values

$$x : y : z : w = aAGHQ : bBHFQ : cCFGQ : abcFGHM;$$

each of the quantities A, B, C, M, Q is linearly expressible in terms of F, G, H , which are themselves connected by the equation $aF + bG + cH = 0$; the foregoing values of x, y, z, w are consequently proportional to quartic functions of a single variable parameter, say $F \div G$; and there is thus an excubo-quartic nodal curve.

To establish the foregoing result, we have

$$\begin{aligned} aA + bH + cG &= 0, \\ aH + bB + cF &= 0, \\ aG + bF + cC &= 0, \\ aA + bB + cC &= 0, \\ aF + bG + cH &= 0, \\ F + G + H &= -M, \\ bcF + caG + abH &= Q, \\ 2bcF &= a^2A - b^2B - c^2C, \\ 2caG &= -a^2A + b^2B - c^2C, \\ 2abH &= -a^2A - b^2B + c^2C, \\ aAGH + bHBF + cFG &= abcM(\alpha + \beta + \gamma)^2, \\ aGH + bHF + cFG &= -abc(\alpha + \beta + \gamma)^2, \\ a^2AGH + b^2HBF + c^2CFG &= -abc\{Q(\alpha + \beta + \gamma)^2 + 3FGH\}, \\ aBCF + bCAG + cABH &= 2Mabc(\alpha + \beta + \gamma)^2, \\ Q(\alpha + \beta + \gamma)^2 + FGH &= ABC, \end{aligned}$$

which are all of them identical equations; but as to some of them the verification is rather complex.

Hence we have

$$\begin{aligned}x + y + z &= Q(aAGH + bBHF + cCFG) \\&= abcMQ(\alpha + \beta + \gamma)^2, \\w &= abcMFGH,\end{aligned}$$

and thence

$$\begin{aligned}x + y + z + w &= abcM\{Q(\alpha + \beta + \gamma)^2 + FGH\} \\&= abcMABC.\end{aligned}$$

Moreover

$$xyzw = (abc)^2 ABC(FGH)^3 Q^3 M,$$

and

$$27a^2b^2c^2xyzw(x + y + z + w)^2 = 27(abc)^6(ABCFGHMQ)^3 (*).$$

Again

$$\begin{aligned}(a^2x + b^2y + c^2z)w &= abcFGHMQ(a^2AGH + b^2BHF + c^2CFG) \\&= (abc)^2FGHMQ\{Q(\alpha + \beta + \gamma)^2 + 3FGH\} \\a^2yz + b^2zx + c^2xy &= abcFGHQ^2(aBCF + bCAG + cABH) \\&= (abc)^2FGHMQ \cdot 2Q(\alpha + \beta + \gamma)^2,\end{aligned}$$

and thence

$$\begin{aligned}a^2(xw + yz) + b^2(yw + zx) + c^2(xy + zw) \\&= 3(abc)^2FGHMQ\{Q(\alpha + \beta + \gamma)^2 + FGH\} \\&= 3(abc)^2ABCFGHMQ \quad (*),\end{aligned}$$

and the two equations marked (*) verify the equation of the surface.

To verify the derived equations, write for a moment $P = a^2(yz + xw) + b^2(zx + yw) + c^2(xy + zw)$, so that the equation of the surface is $P^3 - 27a^2b^2c^2xyzw(x + y + z + w)^2 = 0$, and the derived equation with respect to x is

$$\frac{3}{P} \frac{dP}{dx} = \frac{1}{x} + \frac{2}{x + y + z + w};$$

or substituting for P and $x + y + z + w$ their values, this is

$$\frac{dP}{dx} = \frac{(abc)^2ABCFGHMQ}{x} + 2abcQFGH,$$

and similarly for y , z , and w . In particular, considering the derived equation in respect to w , this is

$$\begin{aligned}a^2x + b^2y + c^2z &= abcABCQ + 2abcQFGH \\&= abcQ(ABC + 2FGH),\end{aligned}$$

and we have as before

$$\begin{aligned}a^2x + b^2y + c^2z &= Q(a^2AGH + b^2BHF + c^2CFG) \\&= abcQ\{\alpha + \beta + \gamma\}^2 + 3FGH \\&= abcQ(ABC + 2FGH),\end{aligned}$$

which is thus verified; the verification of the derived equations for y , z , w can be effected, but not quite so easily.

The existence of the excubo-quartic nodal curve is thus established.

399.

ON THE CUBICAL DIVERGENT PARABOLAS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IX. (1868), pp. 185—189.]

NEWTON reckons five forms, viz. these are the *simplex*, the *complex*, the *crunodal*, the *acnodal*, and the *cuspidal*, but as noticed by Murdoch, the simplex has three different forms, the *ampullate*, the *neutral*, and the *campaniform*. We have thus the 8 forms at once distinguishable by the eye.

Plücker has in all 13 species, the division into species being established or completed geometrically by reference to the asymptotic cuspidal curve (or asymptotic semi-cubical parabola), and analytically as follows, viz. writing the equation in the form

$$y^2 = x^3 - 3cx + 2d,$$

the different species are

simplex,	$y^2 = x^3 - 3cx + 2d$	$\left. \vphantom{\begin{matrix} y^2 = x^3 - 3cx + 2d \\ y^2 = x^3 - 3cx - 2d \end{matrix}} \right\} c^2 < d^2,$	ampullate,
„	$y^2 = x^3 - 3cx - 2d$		campaniform,
„	$y^2 = x^3 + 2d,$		neutral,
„	$y^2 = x^3 - 2d,$		campaniform,
„	$y^2 = x^3 + 3cx + 2d,$		„
„	$y^2 = x^3 + 3cx,$		„
„	$y^2 = x^3 + 3cx - 2d,$		„
complex,	$y^2 = x^3 - 3cx + 2d$	$\left. \vphantom{\begin{matrix} y^2 = x^3 - 3cx + 2d \\ y^2 = x^3 - 3cx - 2d \end{matrix}} \right\} c^2 > d^2,$	
„	$y^2 = x^3 - 3cx - 2d$		
„	$y^2 = x^3 - 3cx,$		
acnodal,	$y^2 = x^3 - 3cx - 2c \sqrt{c},$		
crunodal,	$y^2 = x^3 - 3cx + 2c \sqrt{c},$		
cuspidal,	$y^2 = x^3;$		

but of the simplex species, there are five which are to the eye campaniform, and the three complex species have with each other a close resemblance in form.

I remark as regards the simplex forms, that the tangents at the two inflexions meet in a point R on the axis, and that the ampullate, the neutral, and campaniform forms are distinguished from each other according to the position of R , viz. for the ampullate form, R lies within the curve, for the campaniform form R lies without the curve, and for the neutral form, R is at infinity. It is to be observed, as regards the complex forms, that here R always lies without the curve, between the infinite branch and the oval.

The further division of the simplex and complex forms so as to obtain the $7+3$ species of Plücker, may be effected by considering in conjunction with the point R a certain other point I on the axis; it is to be remarked that excluding the inflexion at infinity the cubical divergent parabola has in all eight inflexions, two real and six imaginary, viz. the inflexions lie by pairs on four ordinates, or if x be the abscissa corresponding to an inflexion, x is determined by a quartic equation; this equation has always two real and two imaginary roots, each of the imaginary roots gives a pair of imaginary inflexions; one of the real roots gives a positive value for y^2 and therefore two real inflexions, the tangents at these meet in the above-mentioned point R on the axis; the other real root gives a negative value for y^2 and therefore two imaginary inflexions, but the tangents at these meet in a real point on the axis, and this I call the point I . It is clear that for each of the four pairs of inflexions the tangents at the two inflexions meet at a point on the axis, so that if X be the abscissa of such point, then X is determined by a quartic equation; two of the roots of this equation are imaginary, the other two roots are real, and correspond to the points R and I respectively.

The equation of the curve being as above

$$y^2 = x^3 - 3cx + 2d,$$

then the coordinate x belonging to a pair of inflexions is found by the equation

$$x^4 - 6cx^2 + 8dx - 3c^2 = 0,$$

or what is the same thing,

$$(1, 0, -c, 2d, -3c^2 \chi x, 1)^4 = 0,$$

(the invariant I is $=0$, and hence the discriminant, $= -27J^2$, is negative, or the roots are two real, two imaginary, as already mentioned): the corresponding value of X is easily found to be

$$X = \frac{x^3 + 3cx - 4d}{3(x^2 - c)},$$

and we thence obtain

$$3cX^4 - 4dX^3 - 6c^2X^2 + 12cdX - (c^3 + 4d^2) = 0,$$

or what is the same thing,

$$(3c, -d, -c^2, 3cd, -c^3 - 4d^2 \chi X, 1)^4 = 0,$$

for the equation in X ; the quadrinvariant I is $=0$, and hence the discriminant, $=-27J^2$, is negative; that is, the roots are two real, two imaginary, as already mentioned.

Considering the simplex forms, first, if $c=0$, then for the curve

$$y^2 = x^3 + 2d,$$

it appears that R lies at infinity, I within the curve; and for the curve

$$y^2 = x^3 - 2d,$$

that R lies without the curve, I at infinity.

It further appears that when $d=0$, or for the curve,

$$y^2 = x^3 + 3cx,$$

R, I lie equidistant from the vertex, R without, I within the curve.

Hence in the curve

$$y^2 = x^3 + 3cx + 2d,$$

since, when $d=0$, the points R, I are equidistant from the vertex, and for $c=0$, the point R is at infinity, it is easy to infer by continuity that the points R, I lie R without, I within the curve, I being nearer to the vertex.

And similarly in the curve

$$y^2 = x^3 + 3cx - 2d,$$

that the points R, I lie R without, I within the curve, R being nearer to the vertex.

Again, in the curve

$$y^2 = x^3 - 3cx + 2d,$$

since, in the curve $y^2 = x^3 + 3cx + 2d$, R is without, I within the curve, and as c becomes $=0$, R passes off to infinity, it appears that c having changed its sign, or for the curve now in question, R having passed through infinity, will be situate within the curve; that is, R, I lie each of them within the curve.

And similarly for the curve

$$y^2 = x^3 - 3cx - 2d,$$

it appears that R, I lie each without the curve.

Hence, finally, for the simplex forms, we have the 7 species of Plücker, viz.

$$y^2 = x^3 - 3cx + 2d, \quad c^3 < d^2,$$

simplex ampullate, R, I within the curve;

$$y^2 = x^3 - 3cx - 2d, \quad c^3 < d^2,$$

simplex campaniform, R, I without the curve;

$$y^2 = x^3 + 2d,$$

simplex neutral, I within the curve, R at infinity;

$$y^2 = x^3 - 2d,$$

simplex campaniform quasi-neutral, R without the curve, I at infinity;

$$y^2 = x^3 + 3cx + 2d,$$

simplex campaniform, R without and further from, I within and nearer to the curve;

$$y^2 = x^3 + 3cx,$$

simplex campaniform equidistant, viz. R and I are equidistant from the curve, R without and I within;

$$y^2 = x^3 + 3cx - 2d,$$

simplex campaniform, R without and nearer to, I within and further from the curve.

Passing to the complex forms, suppose for a moment that α is the diameter of the oval and β the distance of the oval from the vertex of the infinite branch; the equation of the curve then is $y^2 = x(x - \alpha)(x - \alpha - \beta)$, or changing the origin so as to make the term in x^2 to vanish, this is

$$y^2 = (x + \frac{2}{3}\alpha + \frac{1}{3}\beta)(x - \frac{1}{3}\alpha + \frac{1}{3}\beta)(x - \frac{1}{3}\alpha - \frac{2}{3}\beta),$$

or, what is the same thing,

$$y^2 = x^3 - \frac{2}{3}(\alpha^2 + \alpha\beta + \beta^2)x - \frac{1}{27}(\alpha - \beta)(2\alpha + \beta)(\alpha + 2\beta),$$

or comparing this with $y^2 = x^3 - 3cx + 2d$, d is $+$, 0 or $-$, as $\alpha < \beta$, $\alpha = \beta$, $\alpha > \beta$, or say as the oval is smaller, mean, or larger; viz. the magnitude of the oval is estimated by the relation which the diameter thereof bears to the distance of the oval from the infinite branch. In the case $d = 0$, or for the curve $y^2 = x^3 - 3cx$ it appears (as for the corresponding simplex form $y^2 = x^3 + 3cx$) that the points R , I are equidistant from the point $x = 0$, which is in the present case the middle vertex, or vertex of the oval which vertex is nearest to the infinite branch. As the oval diminishes, so that the curve becomes ultimately acnodal, I remaining within the oval ultimately coincides with the acnode; and as the oval increases so that the curve becomes ultimately crunodal, R remaining between the oval and the infinite branch, ultimately coincides with the crunode; and it hence easily appears by continuity that for a smaller oval I is nearer to, R further from the middle vertex; while for a larger oval, I is further from, R nearer to the middle vertex. Hence for the complex forms the species are

$$y^2 = x^3 - 3cx + 2d,$$

smaller oval, I nearer to, R further from the middle vertex;

$$y^2 = x^3 - 3cx,$$

mean oval, R and I equidistant from the middle vertex;

$$y^2 = x^3 - 3cx - 2d,$$

larger oval, I further from, R nearer to the middle vertex: and the division into species is thus completed.

Cambridge, June 16, 1865.

400.

ON THE CUBIC CURVES INSCRIBED IN A GIVEN PENCIL OF SIX LINES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. IX. (1868), pp. 210—221.]

WE have to consider a pencil of six lines, that is, six lines meeting in a point, and a cubic curve touching each of the six lines. As a cubic curve may be made to satisfy nine conditions, the cubic curve will involve three arbitrary parameters; but if we have any particular curve touching the six lines, then transforming the whole figure homologously, the centre of the pencil being the pole and any line whatever the axis of homology, the pencil of lines remains unaltered, and the new curve touches the six lines of the pencil; the transformation introduces three arbitrary constants, and the general solution is thus given as such homologous transformation of a particular solution. To show the same thing analytically, take $(x=0, y=0, z=0)$ for the axes of coordinates, the lines $x=0, y=0$ being any two lines through the centre of the pencil, so that the equation of the pencil is $(\xi x, y)^6=0$, then if $\phi(x, y, z)=0$ is the equation of a cubic curve touching the six lines, the equation of the general curve touching the six lines will be $\phi(x, y, \alpha x + \beta y + \gamma z)=0$; or what is the same thing, considering the coordinate z as implicitly containing three arbitrary constants, viz. an arbitrary multiplier and the two arbitrary parameters of the line $z=0$, then the equation $\phi(x, y, z)=0$ may be taken to be that of the cubic touching the six lines.

Now the given binary sextic $(\xi x, y)^6$ may be expressed in the form $P^2 + Q^3$, where P is a cubic function, Q a quadric function, of the coordinates (x, y) ; or, what is the same thing, but introducing for homogeneity a constant c , we may write

$$(\xi x, y)^6 = c[(a, h, k, b\xi x, y)^3]^2 + 4[(j, l, f\xi x, y)^2]^3;$$

in fact, comparing the two sides of this equation, we have each of the seven coefficients of the sextic equal to a function of the seven quantities $a\sqrt{c}$, $h\sqrt{c}$, $k\sqrt{c}$, $b\sqrt{c}$, j , l , f ; so that conversely, these seven quantities are determinable (not however rationally) in terms of the coefficients of the given sextic. And when the sextic is expressed in the foregoing form, then it will presently be shown that we have

$$(a, h, k, b\sqrt{x}, y)^3 + 3z(j, l, f\sqrt{x}, y)^2 + cz^3 = 0,$$

or, what is the same thing,

$$(a, b, c, f, 0, h, 0, j, k\sqrt{x}, y, z)^3 = 0,$$

as the equation of a cubic curve touching the six given lines; and by what precedes, it appears that this may be taken to be the equation of the general cubic curve which touches the six given lines. On account of the arbitrary constant c , it is sufficient to replace z by $\alpha x + \beta y + z$, or, what is the same thing, to consider $z=0$ as the equation of an arbitrary line, but without introducing therein an arbitrary multiplier.

To sustain the foregoing result, consider the cubic

$$(a, b, c, f, g, h, i, j, k, l\sqrt{x}, y, z)^3 = 0,$$

then in general if $A = (,,\sqrt{x}, y, z)^3$, $B = (,,\sqrt{x}, y, z)^2(\alpha, \beta, \gamma)$, $C = (,,\sqrt{x}, y, z)(\alpha, \beta, \gamma)^2$, $D = (,,\sqrt{\alpha, \beta, \gamma})^3$, the equation of the pencil of tangents drawn from the point (α, β, γ) to the curve is

$$A^2D^2 - 6ABCD + 4AC^3 + 4B^3D - 3B^2C^2 = 0,$$

but writing for shortness

$$(,,\sqrt{x}, y, z)^3 = (A', B', C', D'\sqrt{1}, z)^3,$$

so that

$$A' = (a, h, k, b\sqrt{x}, y)^3,$$

$$B' = (j, l, f\sqrt{x}, y)^2,$$

$$C' = (g, i\sqrt{x}, y),$$

$$D' = c,$$

then for the tangents from the point $(x=0, y=0)$, writing $(\alpha, \beta, \gamma) = (0, 0, 1)$, we have

$$A = (A', B', C', D'\sqrt{1}, z)^3,$$

$$B = (B', C', D'\sqrt{1}, z)^2,$$

$$C = (C', D'\sqrt{1}, z),$$

$$D = D',$$

and thence the equation of the pencil of tangents is

$$A'^2D'^2 - 6A'B'C'D' + 4A'C'^3 + 4B'^3D' - 3B'^2C'^2 = 0.$$

Hence for the curve

$$(a, b, c, f, 0, h, 0, j, k, l\sqrt{x}, y)^3 = 0,$$

we have $g=0$, $i=0$, and therefore $C'=0$; the equation of the pencil of tangents is $A'^2D'^2+4B'^3D'=0$, or throwing out the constant factor D' , and then replacing A' , B' , D' by their values, the equation of the pencil of tangents is

$$c[(a, h, k, b\check{x}, y)^2]^2 + 4[(j, l, f\check{x}, y)^2]^3 = 0,$$

which is the before-mentioned result.

The coefficients $a\sqrt{c}$, $h\sqrt{c}$, $k\sqrt{c}$, $b\sqrt{c}$, j , l , f , or (as we may call them) the coefficients of the cubic curve, are, it has been seen, functions of the coefficients of the given sextic $(\ast\check{x}, y)^6$; hence the invariants S and T of the cubic curve are also functions of the coefficients of the sextic, and it is easy to see that they are in fact invariants (not however rational invariants) of the sextic. To verify this, it is only necessary to show that the invariants S and T are functions of the invariants of the functions $\sqrt{c} \cdot (a, h, k, b\check{x}, y)^3$ and $(j, l, f\check{x}, y)^3$; for if this be so, they will be invariants of the function

$$[c(a, h, k, b\check{x}, y)^3]^2 + 4[(j, l, f\check{x}, y)^3]^3,$$

that is of the sextic. We have in fact the general theorem, that if P , Q , R , ... be any quantics in $(x, y, \dots)$, and $\phi(P, Q, R, \dots)$ a function of these quantics, homogeneous in regard to $(x, y, \dots)$, then any function of the coefficients of ϕ , which is an invariant of the quantics P , Q , R , ... is also an invariant of ϕ .

Considering for greater convenience the function

$$(a, h, k, b\check{x}, y)^3$$

in place of $\sqrt{c} \cdot (a, h, k, b\check{x}, y)^3$, the invariants of the two functions $(a, h, k, b\check{x}, y)^3$ and $(j, l, f\check{x}, y)^3$ are as follows:

$$\begin{aligned} \square &= a^2b^2 - 6ablk + 4ak^3 + 4bl^3 - 3h^2k^2, \\ \nabla &= fj - l^2, \\ \Theta &= j(bh - k^2) + l(hk - ab) + f(ak - h^2), \\ R &= + 1 a^2 f^3 \\ &\quad + 6 abf lj \\ &\quad - 6 ahf^2 l \\ &\quad - 6 akf^2 j \\ &\quad + 12 akf l^2 \\ &\quad + 1 b^2 j^3 \\ &\quad - 6 bhf j^2 \\ &\quad + 12 bhj l^2 \\ &\quad - 6 bkj^2 l \\ &\quad + 9 h^2 f^2 j \\ &\quad - 18 hkf j l \\ &\quad + 9 k^2 f j^2 \\ &\quad - 8 abl^3, \end{aligned}$$

viz. $\square$, ∇ are the discriminants of the two functions respectively, and Θ , R are simultaneous invariants of the two functions, R being in fact the resultant. The corresponding invariants of the functions $\sqrt{c} \cdot (a, h, k, b\check{x}, y)^2$, and $(j, l, f\check{x}, y)^2$ are obviously $c^2\square$, ∇ , $c\Theta$ and cR .

The values of S and T are obtained from the Tables 62 and 63 of my "Third Memoir on Quantics," *Phil. Trans.* vol. CXLVI. (1856), pp. 627—647, [144], by merely writing therein $g=i=0$. It appears that they are in fact functions of $c^2\square$, ∇ , $c\Theta$ and cR ; viz. we have

$$S = \nabla^2 + c\Theta,$$

$$T = 8\nabla^3 + c(4R + 12\nabla\Theta) + c^2\square.$$

The invariants of the sextic $(*\check{x}, y)^6$, if for a moment the coefficients of this sextic are taken to be (a, b, c, d, e, f, g) , that is, if the sextic be represented by $(a, b, c, d, e, f, g\check{x}, y)^6$ are the quadrintvariant $(= ag - 6bf + 15ce - 10d^2)$, Table No. 31 and Salmon's A., p. 203⁽¹⁾, the quartinvariant, No. 34, and Salmon's B., p. 203, the sextinvariant No. 35, and Salmon's C., p. 204, and the discriminant, which is a function of the tenth order $= a^5g^6 + \&c.$ recently calculated for the general form, Salmon, pp. 205—207, say these invariants are Q_2 , Q_4 , Q_6 and Q_{10} . These several invariants are functions of the above-mentioned expressions $c^2\square$, ∇ , $c\Theta$ and cR ; whence, conversely, these quantities are functions of the four invariants Q_2 , Q_4 , Q_6 , Q_{10} ; and the invariants S , T of the cubic curve, being functions of $c^2\square$, ∇ , $c\Theta$ and cR , are also, as they should be, functions of the invariants Q_2 , Q_4 , Q_6 and Q_{10} of the sextic pencil $(*\check{x}, y)^6$.

To effect the calculation of Q_2 , Q_4 and Q_6 , I remark that inasmuch as by a linear transformation, the quadric $(j, l, f\check{x}, y)^2$ may be reduced to the form $2lxy$, and that the invariants of $(a, h, k, b\check{x}, y)^2$ and $2lxy$ are

$$\square = a^2b^2 - 6abhk + 4ak^3 + 4bh^3 - 3h^2k^2,$$

$$\nabla = -l^2,$$

$$\Theta = -l(ab - hk),$$

$$R = -8l^3ab,$$

hence, writing $j=0$, $f=0$, and writing also $c=1$, we may consider the sextic

$$[(a, h, k, b\check{x}, y)^2]^2 + 32l^3x^2y^2,$$

that is

$$(a^2, ah, \frac{1}{5}(2ak + 3h^2), \frac{1}{10}(ab + 9hk + 16l^2), \frac{1}{5}(2bh + 3k^2), bk, b^2\check{x}, y)^6,$$

the invariants whereof are found to be functions of the last mentioned values of $\square$, ∇ , Θ , R ; to pass to the given sextic $(*\check{x}, y)^6$, put equal to

$$c[(a, h, k, b\check{x}, y)^2]^2 + 4[(j, l, f\check{x}, y)^2]^2,$$

we have only to consider $\square$, ∇ , Θ , R as having their before-mentioned general values, and to restore the coefficient c by the principle of homogeneity.

¹ The pages refer to Salmon's *Lessons Introductory to the Modern Higher Algebra* (Second Edition, 1866). In the Fourth Edition, 1885, the values are given, pp. 260—265.

As regards the discriminant Q_{10} , this as already remarked, has been calculated for the general form, but for the present purpose it is easier, by dealing directly with the form $[(a, h, k, b\chi x, y)^3] + 32l^2x^2y^3$, and then interpreting $\square, \nabla, \Theta, R$ and restoring the coefficient c as above, to obtain the discriminant Q_{10} of the function

$$[c(a, h, k, b\chi x, y)^3] + 4[(j, l, f\chi x, y)^2]^3$$

in the required form, as a function of $c^2\square, \nabla, c\Theta, cR$.

I find after some laborious calculations

$$\begin{aligned} Q_2 = 10 \text{ No. 31} &= c^2 \left\{ \begin{array}{l} 9 \square \\ + c \left\{ \begin{array}{l} 40 R \\ + 288 \nabla \Theta \end{array} \right. \\ + \left\{ \begin{array}{l} 256 \nabla^3 \end{array} \right. \end{array} \right. \\ Q_4 = 10000 \text{ No. 34} &= c^4 \left\{ \begin{array}{l} - 99 \square^2 \\ + c^3 \left\{ \begin{array}{l} - 400 R \square \\ + 2304 \nabla \Theta \square \\ + 8640 \Theta^3 \end{array} \right. \\ + c^2 \left\{ \begin{array}{l} 12800 R \nabla \Theta \\ + 82944 \nabla^2 \Theta^2 \\ + 4608 \nabla^3 \square \end{array} \right. \\ + c \left\{ \begin{array}{l} 20480 R \nabla^3 \\ + 147456 \nabla^4 \Theta \end{array} \right. \\ + 65536 \nabla^6 \end{array} \right. \\ Q_6 = 1000000 \text{ No. 35} &= c^6 \left\{ \begin{array}{l} + 7992 \square^3 \\ + c^5 \left\{ \begin{array}{l} + 72000 R \square^2 \\ + 145152 \nabla \Theta^2 \square \\ - 622080 \Theta^3 \square \end{array} \right. \\ + c^4 \left\{ \begin{array}{l} 160000 R^2 \square \\ + 691200 R \nabla \Theta \square \\ + 3456000 R \Theta^3 \\ + 3815424 \nabla^2 \Theta^2 \square \\ + 36080640 \nabla \Theta^4 \\ + 635904 \nabla^3 \square^2 \end{array} \right. \\ + c^3 \left\{ \begin{array}{l} + 33177600 R \nabla^2 \Theta^2 \\ + 4669440 R \nabla^3 \square \\ + 217645056 \nabla^3 \Theta^3 \\ + 23003136 \nabla^4 \Theta \square \end{array} \right. \end{array} \right. \end{aligned}$$

$$\begin{aligned}
& + c^2 \left\{ \begin{aligned} & + 8192000 R^2 \nabla^3 \\ & + 110100480 R \nabla^4 \Theta \\ & + 509607936 \nabla^5 \Theta^2 \\ & + 14155776 \nabla^6 \square \end{aligned} \right. \\
& + c \left\{ \begin{aligned} & 62914560 R \nabla^6 \\ & + 452984832 \nabla^7 \Theta \end{aligned} \right. \\
& + 134217728 \nabla^9
\end{aligned}$$

Q_{10} = multiple of discriminant

$$\begin{aligned}
& = c^7 \left\{ \begin{aligned} & - R^3 \square^2 \\ & + c^6 \left\{ \begin{aligned} & - 8 R^4 \square \\ & - 24 R^3 \nabla \Theta \square \\ & + 64 R^3 \Theta^3 \end{aligned} \right. \\ & + c^5 \left\{ \begin{aligned} & - 16 R^5 \\ & - 96 R^4 \nabla \Theta \\ & + 48 R^4 \nabla^2 \Theta^2 \\ & - 16 R^4 \nabla^3 \square \end{aligned} \right. \\ & + c^4 \left\{ \begin{aligned} & - 64 R^4 \nabla^3, \end{aligned} \right. \end{aligned}
\end{aligned}$$

to which may be joined

$$\begin{array}{l|l}
Q_2^2 = c^4 \left\{ \begin{aligned} & 81 \square^2 \\ & + c^3 \left\{ \begin{aligned} & 720 R \square \\ & + 5184 \nabla \Theta \square \end{aligned} \right. \\ & + c^2 \left\{ \begin{aligned} & 1600 R^2 \\ & + 23040 R \nabla \Theta \\ & + 82944 \nabla^2 \Theta^2 \\ & + 4608 \nabla^3 \square \end{aligned} \right. \\ & + c \left\{ \begin{aligned} & 20480 R \nabla^3 \\ & + 147456 \nabla^4 \Theta \end{aligned} \right. \\ & + 65536 \square^6 \end{aligned} \right. & Q_2^2 - Q_4 = c^4 \left\{ \begin{aligned} & 180 \square^2 \\ & + c^3 \left\{ \begin{aligned} & 1120 R \square \\ & + 2880 \nabla \Theta \square \\ & - 8640 \Theta^3 \end{aligned} \right. \\ & + c^2 \left\{ \begin{aligned} & 1600 R^2 \\ & + 10240 R \nabla \Theta \end{aligned} \right. \end{aligned} \right.
\end{array}$$

and

$$\begin{aligned}
\frac{1}{8} (Q_6 - 8 Q_2^3) & = c^6 \left\{ \begin{aligned} & 270 \square^3 \\ & + c^5 \left\{ \begin{aligned} & - 720 R \square^2 \\ & - 51840 \nabla \Theta \square^2 \\ & - 77760 \Theta^3 \square \end{aligned} \right. \end{aligned} \right.
\end{aligned}$$

$$\begin{aligned}
& + c^4 \left\{ \begin{array}{l} - 23200 R^2 \square \\ - 535680 R \Theta \nabla \square \\ + 432000 R \Theta^3 \\ - 1762560 \nabla^2 \Theta^2 \square \\ + 4510080 \nabla \Theta^4 \\ + 17280 \nabla^3 \square^2 \end{array} \right. \\
& + c^3 \left\{ \begin{array}{l} - 64000 R^3 \\ - 1382400 R^2 \nabla \Theta \\ - 5806080 R \nabla^2 \Theta^2 \\ + 30720 R \nabla^3 \square \\ + 3317760 \nabla^3 \Theta^3 \\ - 1105920 \nabla^4 \Theta \square \end{array} \right. \\
& + c^2 \left\{ \begin{array}{l} - 204800 R^2 \nabla^3 \\ - 3932160 R \nabla^4 \Theta. \end{array} \right.
\end{aligned}$$

The foregoing values of S and T give

$$\begin{aligned}
T^2 - 64S^3 = & c^4 \left\{ \begin{array}{l} \square^2 \\ + c^3 \left\{ \begin{array}{l} 8 R \square \\ + 24 \nabla \Theta \square \\ - 64 \Theta^3 \end{array} \right. \\ + c^2 \left\{ \begin{array}{l} 16 R^2 \\ + 96 R \nabla \Theta \\ - 48 \nabla^2 \Theta^4 \\ + 16 \nabla^3 \square \end{array} \right. \\ + c \quad 64 R \nabla^3, \end{array} \right.
\end{aligned}$$

so that

$$Q_{10} = -c^3 R^3 (T^2 - 64S^3),$$

$$64S^3 - T^2 = \frac{Q_{10}}{c^3 R^3},$$

$$S = \nabla^3 + c\Theta,$$

and therefore

$$64 - \frac{T^2}{S^3} = \frac{Q_{10}}{c^3 R^3 (\nabla^3 + c\Theta)^3} = \frac{Q_{10}}{[cR (\nabla^3 + c\Theta)]^3},$$

formulae which are interesting in the theory.

We have

$$c\Theta = S - \nabla^3,$$

$$c^2\Box = T - 12\nabla S + 4\nabla^3 - 4cR,$$

and if by means of these values we eliminate $c\Theta$ and $c^2\Box$, we obtain Q_2 , Q_4 , Q_6 and Q_{10} as functions of S , T , ∇ and cR . Choosing instead of Q_4 and Q_6 the combinations $Q_2^2 - Q_4$ and $Q_6 - 8Q_2^3$, and forming also the expression for the combination $Q_2(Q_2^2 - Q_4)$, we have thus the system of formulæ

$Q_2 = 9T + 180\nabla S + 4\nabla^3 + 4cR,$ $\frac{1}{20}(Q_2^2 - Q_4) = + 9T^2$ $- 432S^3$ $- 72T\nabla S$ $- 72T\nabla^3$ $- 16TcR$ $+ 864\nabla^2S^2$ $+ 144\nabla^4S$ $+ 128\nabla ScR,$ $\frac{1}{80}(Q_6 - 8Q_2^3) =$ $+ 27T^3$ $- 4212T^2\nabla S$ $+ 6588T^2\nabla^3$ $+ 252T^2cR$ $- 7776TS^3$ $- 16848T\nabla^2S^3$ $+ 3456T\nabla^4S$ $- 2592T\nabla ScR$ $- 1296T\nabla^6$ $- 1824T\nabla^3cR$ $- 448Tc^2R^2$ $+ 544320\nabla S^4$ $- 461376\nabla^3S^3$ $+ 74304cRS^3$ $+ 15552\nabla^5S^2$ $- 10368\nabla^2S^2cR$ $- 10728\nabla^7S$ $- 3264\nabla^4ScR$ $- 1536\nabla Sc^2R^2,$	$\frac{1}{20}Q_2(Q_2^2 - Q_4) =$ $+ 81T^3$ $+ 972T^2\nabla S$ $- 612T^2\nabla^3$ $- 108T^2cR$ $- 3888TS^3$ $- 5184T\nabla^2S^3$ $- 11952T\nabla^4S$ $- 2016T\nabla ScR$ $- 288T\nabla^6$ $- 352T\nabla^3cR$ $- 64Tc^2R^2$ $- 77760\nabla S^4$ $+ 153792\nabla^3S^3$ $- 1728cRS^3$ $+ 29376\nabla^5S^2$ $+ 26996\nabla^2S^2cR$ $+ 576\nabla^7S$ $+ 1088\nabla^4ScR$ $+ 512\nabla Sc^2R^2,$
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and, as mentioned above,

$$Q_{10} = c^3 R^3 (-T^2 + 64S^3).$$

The just-mentioned value of Q_{10} should, I think, admit of being established *a priori*, and if this be so, then the substitution of the values of S and T in terms of $c^2\Box$, ∇ , $c\Theta$, cR , would be the easiest way of arriving at the before-mentioned expression of Q_{10} in terms of these same quantities. The calculation by which this expression was arrived at, is however not without interest, and it will be as well to indicate the mode in which it was effected.

Calculation of Q_{10} .

We have to find the discriminant of

$$c[(a, h, k, b\chi x, y)^3] + 32l^3x^2y^2.$$

Consider for a moment the more general form $P^2 + 4Q^3$, then to find the discriminant, we have to eliminate between the equations

$$P \frac{dP}{dx} + 6Q^2 \frac{dQ}{dx} = 0,$$

$$P \frac{dP}{dy} + 6Q^2 \frac{dQ}{dy} = 0,$$

these are satisfied by the system $P=0$, $Q^2=0$, and it follows that if R be the resultant of the equations $P=0$, $Q=0$, then the discriminant in question contains the factor R^2 . For the other factor we may reduce the system to

$$P \frac{dP}{dx} + 6Q^2 \frac{dQ}{dx} = 0,$$

$$\frac{dP}{dx} \frac{dQ}{dy} - \frac{dP}{dy} \frac{dQ}{dx} = 0.$$

Now writing $Q = 2lxy$, these equations become

$$P \frac{dP}{dx} + 48l^3x^2y^3 = 0,$$

$$l \left(x \frac{dP}{dx} - y \frac{dP}{dy} \right) = 0,$$

the resultant of which is $= l^3$ into resultant of the system

$$P \frac{dP}{dx} + 48l^3x^2y^3 = 0,$$

$$x \frac{dP}{dx} - y \frac{dP}{dy} = 0,$$

but in virtue of the second equation, we have

$$P = \frac{1}{3} \left(x \frac{dP}{dx} + y \frac{dP}{dy} \right) = \frac{2}{3} y \frac{dP}{dy},$$

which reduces the first equation to

$$2y \frac{dP}{dx} \frac{dP}{dy} + 144l^3x^2y^3 = 0,$$

or omitting the factor $2y$, to

$$\frac{dP}{dx} \frac{dP}{dy} + 72l^3x^2y^3 = 0.$$

Hence, writing $P = \sqrt[3]{(c) \cdot (a, h, k, b\check{x}x, y)^3}$, and therefore $\frac{dP}{dx} = 3 \sqrt[3]{(c) \cdot (a, h, k\check{x}x, y)^2}$, $\frac{dP}{dy} = 3 \sqrt[3]{(c) \cdot (h, k, b\check{x}x, y)^2}$; writing also $y = 1$, the two equations become

$$c(a, h, k\check{x}x, 1)^2(h, k, b\check{x}x, 1)^2 + 8l^3x^2 = 0,$$

$$(a, h, k\check{x}x, 1)^2x - (h, k, b\check{x}x, 1)^2 = 0,$$

the second of which is more simply written

$$(a, h, -k, -b\check{x}x, 1)^3 = 0.$$

Hence, restoring the factor l^3 , and also to avoid fractions introducing the factor $8a^4$, the resultant of the two equations is

$$= 8l^3a^4\Pi \{8l^3x^2 + c(a, h, k\check{x}x, 1)^2(h, k, b\check{x}x, 1)^2\},$$

where Π denotes the product of the factors corresponding to the three roots x_1, x_2, x_3 of the equation

$$(a, h, -k, -b\check{x}x, 1)^3 = 0,$$

or what is the same thing,

$$ax^3 + hx^2 - kx - b = 0,$$

so that the symmetric functions are to be found from

$$\Sigma x_1 = -\frac{h}{a}, \quad \Sigma x_1x_3 = -\frac{k}{a}, \quad x_1x_2x_3 = \frac{b}{a}.$$

The required discriminant is the foregoing resultant multiplied by R , or say by c^2R^3 , that is the discriminant Q_{10} is

$$= c^2R^3 \cdot 8l^3a^4\Pi (8l^3x^2 + \Omega),$$

if for shortness we write

$$\Omega = (a, h, k\check{x}x, 1)^2 \cdot (a, k, b\check{x}x, 1)^2,$$

and when the symmetric functions have been expressed in terms of the coefficients, the result is to be expressed as a function of $\square$, ∇ , Θ , R by means of the values

$$c^2\square = a^2b^2 + 4ak^3 + 4bh^3 - 6adhk - 3h^2k^2,$$

$$\nabla = -l^2,$$

$$c\Theta = -l(ab - hk),$$

$$cR = -8l^3ab.$$

Thus, for instance, the first term of the result is

$$= c^2R^2 \cdot 8l^3a^4 \cdot 512l^3x_1^2x_2^2x_3^2,$$

which is

$$= c^2R^2 \cdot 4096l^{12}a^2b^2,$$

$$= c^2R^2 \cdot -64c^2R^2\nabla^2,$$

$$= -64c^4R^4\nabla^2,$$

which is a term in the before-mentioned expression for Q_{10} .

401.

A NOTATION OF THE POINTS AND LINES IN PASCAL'S THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. ix. (1868), pp. 268—274.]

TAKING six points 1, 2, 3, 4, 5, 6 on a conic; let $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O$ denote each a combination of three lines, thus

$$\begin{array}{|l|l|l|} \hline 12.34.56 = A & 12.35.64 = F & 12.36.45 = K \\ \hline 13.45.62 = B & 13.46.25 = G & 13.42.56 = L \\ \hline 14.56.23 = C & 14.52.36 = H & 14.53.62 = M \\ \hline 15.62.34 = D & 15.63.42 = I & 15.64.23 = N \\ \hline 16.23.45 = E & 16.24.53 = J & 16.25.34 = O \\ \hline \end{array}$$

then any hexagon formed with the six points may be represented by a combination of some two of the letters A, B , &c., viz. the three alternate sides are the lines represented by one letter, and the other three alternate sides the lines represented by the other letter: for example, the hexagon 123456 is AE ; and so for the other hexagons. Any duad AE thus representing a hexagon may be termed a hexagonal duad; the number of such duads is sixty. Each Pascalian line may be denoted by the symbol of the hexagon to which it belongs; thus, the line which belongs to the hexagon AE , is the line AE .

I form the following combinations:

$IMO.DHJ$	each involving all the duads 12, &c. except those of	123.456,
$DEG.BNO$	"	124.356,
$ELM.BCJ$	"	125.346,
$HLN.CGI$	"	126.345,
$EFI.JKN$	"	134.256,
$AEH.CKO$	"	135.246,
$AMN.CDF$	"	136.245,
$AGJ.ELO$	"	145.236,
$ABI.DKL$	"	146.235,
$GKM.BFH$	"	156.234,

and also the combinations:

<i>AEGMI</i>	involving all the duads 12, 13, &c.,
<i>ABHJN</i>	" "
<i>BCFIO</i>	" "
<i>CDGJK</i>	" "
<i>DEFHL</i>	" "
<i>KLMNO</i>	" "

which I call respectively the ten-partite and six-partite arrangements. It is to be remarked that (considering *IMO.DHJ* as standing for the six duads *IM*, *IO*, *MO*, *DH*, *DJ*, *HJ*, and so for the others) the ten-partite arrangement contains all the sixty hexagonal duads: and in like manner, (considering *AEGMI* as standing for the ten duads *AE*, *AG*, *AM*, *AI*, *EG*, *EM*, *EI*, *GM*, *GI*, *MI*, and so for the others) the six-partite arrangement contains all the sixty hexagonal duads.

The 60 Pascalian lines intersect by 4's in the 45 Pascalian points *p*, by 3's in 20 points *g* and in 60 points *h*, and by 2's in 90 points *m*, 360 points *r*, 360 points *t*, 360 points *z*, and 9 points *w*.

The intersections of the Pascalian lines thus are

45 <i>p</i>	counting as	270
20 <i>g</i>	" "	60
60 <i>h</i>	" "	180
90 <i>m</i>	" "	90
360 <i>r</i>	" "	360
360 <i>t</i>	" "	360
360 <i>z</i>	" "	360
90 <i>w</i>	" "	90
		1770 = $\frac{1}{2}60 \cdot 59$,

and the intersections on each Pascalian line are

3 <i>p</i>	counting as	9
1 <i>g</i>	" "	2
3 <i>h</i>	" "	6
3 <i>m</i>	" "	3
12 <i>r</i>	" "	12
12 <i>t</i>	" "	12
12 <i>z</i>	" "	12
3 <i>w</i>	" "	3
		59.

For the ten-partite arrangement, any double triad such as $ABI.DKL$ gives 15 intersections; $10 \times 15 = 150$; and any pair of double triads such as $ABI.DKL$ and $AEH.CKO$ gives 36 intersections; $45 \times 36 = 1620$; and these are

$$\begin{array}{rcl}
 10 \times \left\{ \begin{array}{l} 6g \\ 9m \\ \hline 15 \end{array} \right. & \begin{array}{r} 60g \\ 90m \\ \hline 150 \end{array} \\
 45 \times \left\{ \begin{array}{l} 6p \\ 4h \\ 8r \\ 8t \\ 8z \\ 2w \\ \hline 36 \end{array} \right. & \begin{array}{r} 270p \\ 180h \\ 360r \\ 360t \\ 360z \\ 90w \\ \hline 1620 \\ 1770. \end{array}
 \end{array}$$

For the six-partite arrangement any pentad such as $ABHJN$ gives 45 intersections; $6 \times 45 = 270$; and any two pentads such as $ABHJN$ and $AEGMI$ give 100 intersections; $15 \times 100 = 1500$; and these are

$$\begin{array}{rcl}
 6 \times \left\{ \begin{array}{l} 30h \\ 15m \\ \hline 45 \end{array} \right. & \begin{array}{r} 180h \\ 90m \\ \hline 270 \end{array} \\
 15 \times \left\{ \begin{array}{l} 4g \\ 18p \\ 24r \\ 24t \\ 24z \\ 6w \\ \hline 100 \end{array} \right. & \begin{array}{r} 60g \\ 270p \\ 360r \\ 360t \\ 360z \\ 90w \\ \hline 1500 \\ 1770. \end{array}
 \end{array}$$

I analyse the intersections of a Pascalian line, say AE , by the remaining 59 Pascalian lines as follows:

Observe that AE belongs to the triad AEH , the complementary triad whereof is CKO ; it also belongs to the pentad $AEIMG$. We thus obtain, corresponding to AE , the arrangement

$$\begin{array}{c}
 H \ H \ H \\
 H \ A \ \boxed{\begin{array}{c} B \ N \ J \\ F \ L \ D \end{array}} \\
 H \ E \ \boxed{\begin{array}{c} I \ M \ G \\ K \ C \ O \end{array}}
 \end{array}$$

viz. $HABNJ$, is the pentad which contains HA , the arrangement of the last three letters B, N, J thereof being arbitrary; $HEFLD$ is the pentad that contains HE , but the last three letters are so arranged that the columns HB, HF, HN, HJ, HD are each of them a triad, IMG is then the residue of the pentad $AEIMG$, and KCO is the complementary triad to AEH , but the arrangement of the letters IMG , and of the letters KCO , are each of them determinate; viz. these are such that we have $BFICO$, $NLMKO$, $JDGCK$, each of them a pentad.

And this being so we derive from the arrangement

$$\begin{array}{l}
 2g \quad AH, EH; \\
 3m \quad KC, KO, CO; \\
 6h \quad AI, AM, AG; EI, EM, EG; \\
 12z \quad IB, IF, MN, ML, GJ, GD; HB, HF, HN, HL, HJ, HD; \\
 9p \quad AB, AN, AJ; EF, EL, ED; BF, NL, JD; \\
 12r \quad CB, CF, CJ, CD; OB, OF, ON, OL; KN, KL, KJ, KD; \\
 12t \quad FL, FD, LD; BN, BJ, NJ; IC, IO; MK, MO; GK, GC; \\
 3w \quad IM, IG, MG; \\
 \hline
 59
 \end{array}$$

viz. the line AE in question meets AH, EH each of them in a point g ; KC, KO, CO each in a point m ; and so on. By constructing in the same way an arrangement for each of the lines AH , &c., we find the nature of the point of intersection of any two of the lines AB, AE, AH , &c.; and we may then present the results in a table (see Plate), which shows at a glance what is the point of intersection (whether a point g, m, h, z, p, r, t , or w) of any two of the Pascalian lines.

I further remark that representing the 45 Pascalian points as follows:

12.34 = a	13.24 = g	14.23 = m	15.23 = s	16.23 = y
12.35 = b	13.25 = h	14.25 = n	15.24 = t	16.24 = z
12.36 = c	13.26 = i	14.26 = o	15.26 = u	16.25 = α
12.45 = d	13.45 = j	14.35 = p	15.34 = v	16.34 = β
12.46 = e	13.46 = k	14.36 = q	15.36 = w	16.35 = γ
12.56 = f	13.56 = l	14.56 = r	15.46 = x	16.45 = δ
23.45 = ϵ	25.34 = λ	34.56 = ρ		
23.46 = ζ	25.36 = μ	35.46 = σ		
23.56 = η	25.46 = ν	36.45 = τ		
24.35 = θ	26.34 = ξ			
24.36 = ι	26.35 = ω			
24.56 = κ	26.45 = π			

the sixty hexagons and their Pascalian lines then are

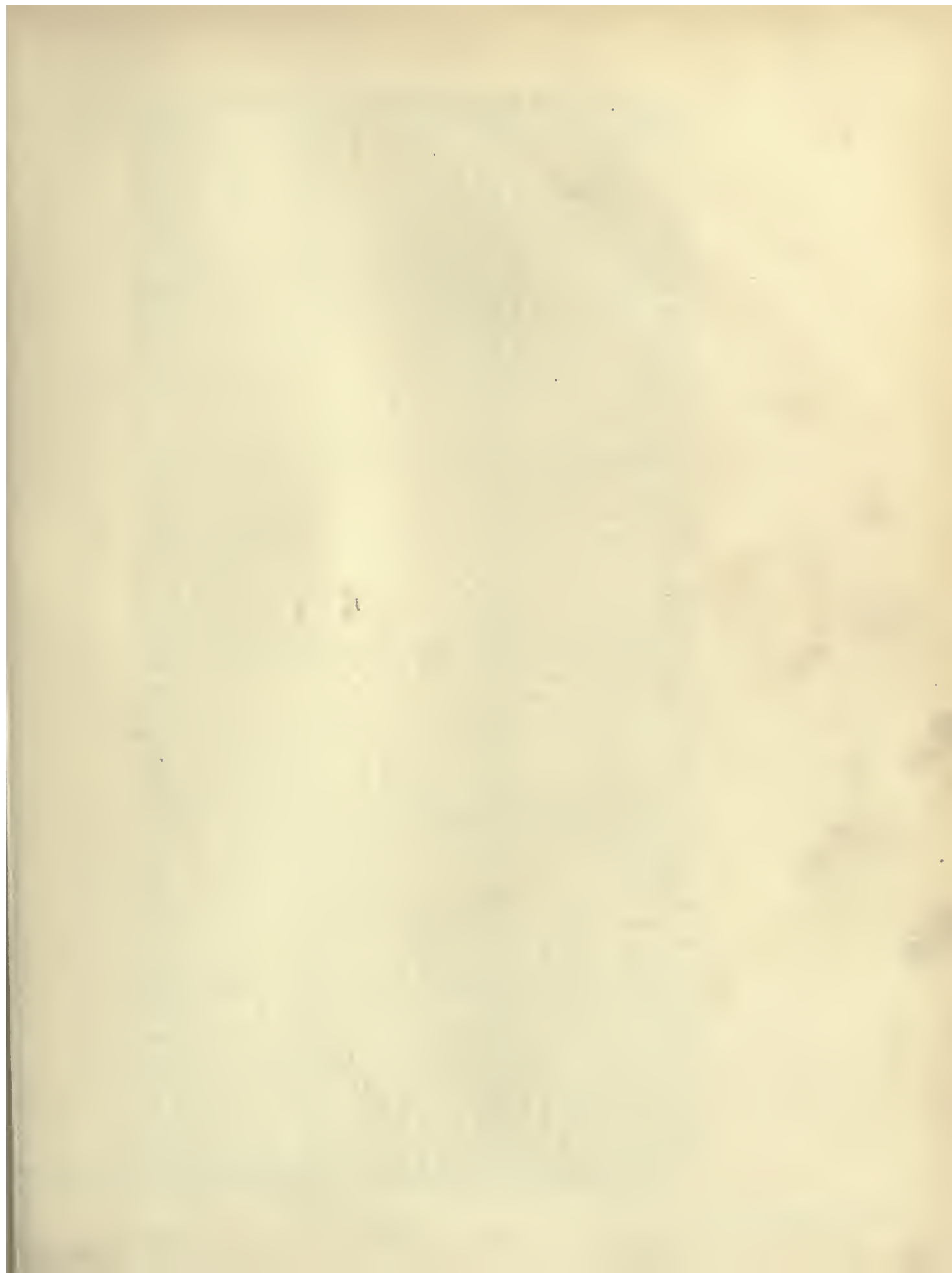
<i>AE</i>	123456	12.45	23.56	34.61	$d\eta\beta$
<i>AH</i>	125634	12.63	25.34	56.41	$c\lambda r$
<i>EH</i>	145236	14.23	45.36	52.61	$m\tau\alpha$
<i>CK</i>	123654	12.65	23.54	36.41	$f\epsilon q$
<i>CO</i>	143256	14.25	43.56	32.61	$n\pi y$
<i>KO</i>	125436	12.43	25.36	54.61	$a\mu\delta$
<i>AM</i>	126534	12.53	26.34	65.41	$b\xi r$
<i>AG</i>	125643	12.64	25.43	56.31	$e\lambda l$
<i>AI</i>	124365	12.36	24.65	43.51	$c\kappa v$
<i>EG</i>	132546	13.54	32.46	25.61	$j\zeta\alpha$
<i>DF</i>	126435	12.43	26.35	64.51	$a\omega x$
<i>FL</i>	124653	12.65	24.53	46.31	$f\theta k$
<i>DL</i>	134265	13.26	34.65	42.51	$i\rho t$
<i>BN</i>	132645	13.64	32.45	26.51	$k\epsilon u$
<i>BJ</i>	135426	13.42	35.26	54.61	$g\omega\delta$
<i>JN</i>	153246	15.24	53.46	32.61	$t\sigma y$
<i>GK</i>	125463	12.46	25.63	54.31	$e\mu j$
<i>KM</i>	126354	12.35	26.54	63.41	$b\pi q$
<i>IO</i>	152436	15.43	52.36	24.61	$v\mu z$
<i>MO</i>	143526	14.52	43.26	35.61	$n\xi\gamma$
<i>EM</i>	145326	14.32	45.26	53.61	$m\pi\gamma$
<i>EI</i>	154236	15.23	54.36	42.61	$s\tau z$
<i>AN</i>	123465	12.46	23.65	34.51	$e\eta v$
<i>AJ</i>	124356	12.35	24.56	43.61	$b\kappa\beta$
<i>AB</i>	126543	12.54	26.43	65.31	$d\zeta l$
<i>DE</i>	154326	15.32	54.26	43.61	$s\pi\beta$
<i>EL</i>	132456	13.45	32.56	24.61	$j\eta z$
<i>EF</i>	123546	12.54	23.46	35.61	$d\xi\gamma$
<i>CD</i>	143265	14.26	43.65	32.51	$\rho\sigma s$
<i>CF</i>	123564	12.56	23.64	35.41	$f\zeta p$

<i>CG</i>	132564	13.56	32.64	25.41	$l\zeta n$
<i>CI</i>	142365	14.36	42.65	23.51	$q\kappa s$
<i>MN</i>	146235	14.23	46.35	62.51	$m\sigma u$
<i>GJ</i>	135246	13.24	35.46	52.61	$g\sigma\alpha$
<i>BI</i>	136245	13.24	36.45	62.51	$g\tau u$
<i>DG</i>	134625	13.62	34.25	46.51	$i\lambda x$
<i>LM</i>	135624	13.62	35.24	56.41	$i\theta r$
<i>FI</i>	124635	12.63	24.35	46.51	$c\theta x$
<i>BH</i>	136254	13.25	36.54	62.41	$h\tau o$
<i>FH</i>	125364	12.36	25.64	53.41	$c\nu p$
<i>FO</i>	125346	12.34	25.46	53.61	$a\nu\gamma$
<i>LO</i>	134256	13.25	34.56	42.61	$h\rho z$
<i>DK</i>	126345	12.34	26.45	63.51	$a\pi w$
<i>KL</i>	124563	12.56	24.63	45.31	$f\iota j$
<i>BO</i>	134526	13.52	34.26	45.61	$h\xi\delta$
<i>NO</i>	152346	15.34	52.46	23.61	$v\nu y$
<i>BC</i>	132654	13.65	32.54	26.41	$l\epsilon o$
<i>CJ</i>	142356	14.35	42.56	23.61	$p\kappa y$
<i>JK</i>	124536	12.53	24.36	45.61	$b\iota\delta$
<i>KN</i>	123645	12.64	23.45	36.51	$e\epsilon w$
<i>DH</i>	143625	14.62	43.25	36.51	$o\lambda w$
<i>HJ</i>	142536	14.53	42.36	25.61	$p\iota\alpha$
<i>HL</i>	136524	13.52	36.24	65.41	$h\iota r$
<i>HN</i>	146325	14.32	46.25	63.51	$m\nu w$
<i>BF</i>	126453	12.46	26.53	64.31	$d\omega k$
<i>DJ</i>	153426	15.42	53.26	34.61	$t\omega\beta$
<i>LN</i>	132465	13.46	32.65	24.51	$k\eta t$
<i>GM</i>	135264	13.26	35.64	52.41	$i\sigma n$
<i>IM</i>	142635	14.63	42.35	26.51	$q\theta u$
<i>GI</i>	136425	13.42	36.25	64.51	$g\mu x$

Each Pascalian point belongs to four different hexagons; viz. a to the hexagons KD, KO, FD, FO ; and so for the other points, thus:

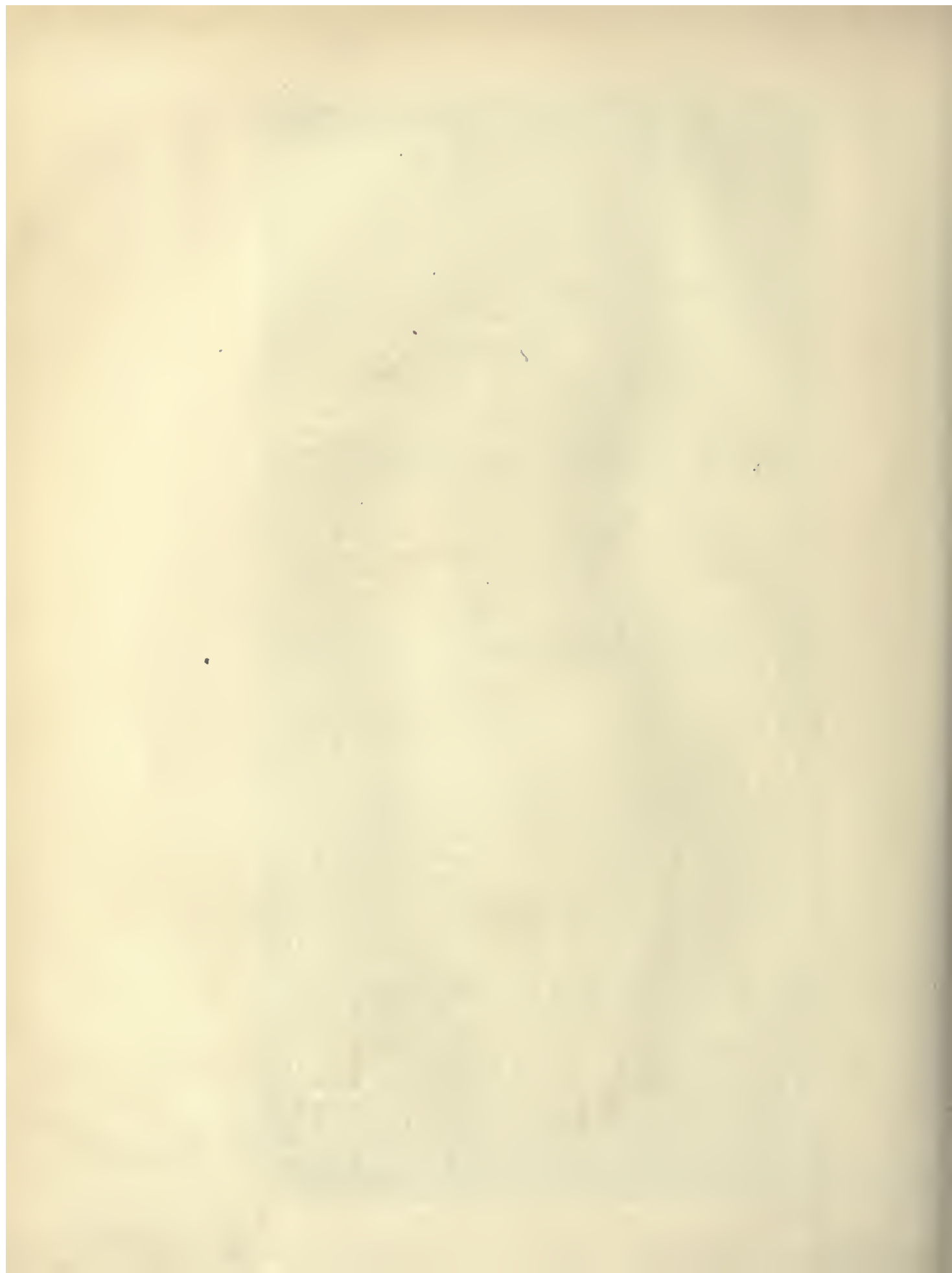
a	$(K, F)(D, O)$	x	$(D, I)(F, G)$
b	$(A, K)(M, J)$	y	$(C, N)(J, O)$
c	$(A, F)(H, I)$	z	$(E, O)(I, L)$
d	$(A, F)(B, E)$	α	$(E, J)(G, H)$
e	$(A, K)(G, N)$	β	$(A, D)(E, J)$
f	$(C, L)(K, F)$	γ	$(E, O)(F, M)$
g	$(B, G)(I, J)$	δ	$(B, K)(J, O)$
h	$(B, L)(H, O)$	ϵ	$(B, K)(C, N)$
i	$(D, M)(G, L)$	ζ	$(C, E)(F, G)$
j	$(E, K)(G, L)$	η	$(A, L)(E, N)$
k	$(B, L)(F, N)$	θ	$(F, M)(I, L)$
l	$(A, C)(B, G)$	ι	$(H, K)(J, L)$
m	$(E, N)(H, M)$	κ	$(A, C)(I, J)$
n	$(C, M)(G, O)$	λ	$(A, D)(G, H)$
o	$(B, D)(C, H)$	μ	$(G, O)(I, K)$
p	$(C, H)(F, J)$	ν	$(F, N)(H, O)$
q	$(C, M)(I, K)$	ξ	$(A, O)(B, M)$
r	$(A, L)(H, M)$	ω	$(B, D)(F, J)$
s	$(C, E)(D, I)$	π	$(D, M)(E, K)$
t	$(J, L)(D, N)$	ρ	$(C, L)(D, O)$
u	$(B, M)(I, N)$	σ	$(G, N)(J, M)$
v	$(A, O)(N, I)$	τ	$(B, E)(H, I)$
w	$(D, N)(H, K)$		

I have constructed on a very large scale a figure of the sixty Pascalian lines, and the forty-five Pascalian points, marking them according to the foregoing notation; but the figure is from its complexity, and the inconvenient way in which the points are either crowded together or fly off to a great distance, almost unintelligible.



	AB	AE	AG	AH	AI	AJ	AM	AN	BC	BF	BH	BI	BJ	BN	BO	CD	CF	CG	CI	CJ	CK	CO	DE	DF	DG	DH	DJ	DK	DL	EF
AB		p	p	h	g	h	p	h	p	p	h	g	h	p	r	t	p	x	x	r	t	r	r	r	t	t	m	m	p	
AE	p		h	g	h	p	h	p	r	p	x	x	t	t	r	r	r	t	t	r	m	m	p	t	x	x	p	r	t	p
AG	p	h		p	h	g	h	p	p	r	t	x	x	t	r	t	r	p	x	x	t	r	x	r	p	p	x	t	r	t
AH	h	g	p		p	h	p	h	t	x	h	x	w	t	r	r	r	r	t	m	m	x	t	p	p	x	r	t	x	
AI	g	h	h	p		p	h	p	x	x	g	x	x	x	r	t	x	p	p	r	t	t	r	t	r	m	m	x		
AJ	h	p	g	h	p		p	h	x	t	w	x	h	w	t	t	r	x	p	p	t	r	p	r	x	x	p	t	r	r
AM	p	h	h	p	h	p		g	r	r	t	x	t	x	p	m	m	t	t	r	r	r	t	m	t	r	r	r	t	
AN	h	p	p	h	p	h	g		t	t	w	x	w	h	x	m	m	r	r	t	r	r	r	m	r	t	t	r	r	r
BC	p	r	p	t	x	x	r	t		h	p	h	g	p	h	p	h	p	h	g	p	h	r	x	t	p	x	t	r	t
BF	p	p	r	x	x	t	r	t	h		g	h	p	p	h	x	h	t	w	x	t	w	t	p	r	x	p	r	t	p
BH	h	x	t	h	x	w	t	w	p	g		p	h	h	p	p	x	r	t	x	r	t	t	x	r	p	x	r	t	x
BI	g	x	x	g	x	x	x	h	h	p		p	p	h	t	w	x	h	x	l	w	r	t	r	r	r	m	m	x	
BJ	h	t	x	w	x	h	t	w	g	p	h	p		h	p	x	x	x	g	x	x	r	p	t	x	p	t	r	r	
BN	h	t	t	w	x	w	x	h	p	p	h	p	h		g	r	t	r	t	x	p	x	m	r	m	t	t	r	r	r
BO	p	r	r	t	x	t	p	x	h	h	p	h	p	g		t	w	t	w	x	x	h	m	t	m	r	r	r	r	t
CD	r	r	t	r	r	t	m	m	p	x	p	t	x	r	t		g	h	p	h	p	p	g	h	p	h	p	x		
CF	t	r	r	r	t	r	m	m	h	h	x	w	x	t	w	g		p	h	p	p	h	x	g	x	x	x	x	p	
CG	p	t	p	r	x	t	r	p	t		r	x	x	r	t	h	p		g	h	p	x	x	h	t	w	w	t	p	
CI	x	t	x	r	p	p	t	r	h	w	t	h	x	t	w	p	h	g		p	p	h	p	x	x	r	t	t	r	x
CJ	x	r	x	t	p	p	r	t	g	x	x	g	x	x	h	p	h	p		h	p	t	x	w	x	h	w	t	r	
CK	r	m	t	m	r	t	r	r	p	t	r	t	x	p	x	h	p	h	p	h		g	t	x	w	t	w	h	x	r
CO	t	m	r	m	t	r	r	r	h	w	t	w	x	x	h	p	h	p	h	p	g		r	x	t	r	t	x	p	t
DE	r	p	x	x	t	p	t	r	r	t	t	r	r	m	m	p	x	x	p	t	t	r		h	g	h	p	p	h	h
DF	r	t	r	t	r	r	m	m	x	p	x	t	p	r	t	g	g	x	x	x	x	h		p	h	p	p	h	h	
DG	r	x	p	p	t	x	t	r	t	r	r	r	t	m	m	h	x	h	x	w	w	t	g	p		p	h	p	x	
DH	t	x	p	p	r	x	r	t	p	x	p	r	x	t	r	p	x	t	r	x	t	r	h	h	p		g	p	h	w
DJ	t	p	x	x	r	p	r	t	x	p	x	r	p	t	r	h	x	w	t	h	w	t	p	h	h	g		h	p	t
DK	m	r	t	r	m	t	r	r	t	r	r	m	t	r	r	h	x	w	t	w	h	x	p	p	h	p	h		g	t
DL	m	t	r	t	m	r	r	r	r	t	t	m	r	r	r	p	x	t	r	t	x	p	h	h	p	h	p	g		w
EF	p	p	t	x	x	r	t	r	t	p	x	x	r	r	t	x	p	p	x	r	r	t	h	h	x	w	t	t	w	
EG	t	h	h	x	w	x	w	t	r	r	r	t	r	m	m	x	p	p	x	t	t	r	g	x	g	x	x	x	p	
EH	x	g	x	g	x	x	x	r	x	p	p	t	t	r	t	t	r	r	r	m	m	h	w	x	h	x	t	w	h	
EI	x	h	w	x	h	t	w	t	t	x	p	p	r	r	t	p	x	x	p	r	r	t	p	x	r	t	r	t	g	
EL	r	p	t	x	t	r	x	p	m	t	t	r	m	r	r	t	t	r	m	r	r	h	w	x	w	t	x	h	h	
EM	t	h	w	x	w	t	h	x	m	r	r	t	m	r	r	r	r	t	t	m	r	r	p	t	x	t	r	p	x	p
FI	x	x	t	p	p	r	t	r	w	h	x	h	t	t	w	x	h	x	h	t	t	w	x	p	p	t	r	r	t	g
FH	x	x	r	p	p	t	r	t	x	g	g	x	x	x	x	p	r	t	p	r	t	w	h	t	h	x	t	w	h	
FL	r	t	m	t	r	m	r	r	t	p	x	t	r	p	x	p	r	t	r	p	x	w	h	t	w	t	x	h	h	
FO	t	r	m	r	t	m	r	r	w	h	x	w	t	x	h	x	h	t	w	t	x	h	t	p	r	t	r	p	x	p
GI	x	w	h	t	h	x	w	t	x	t	r	p	p	r	t	x	g	g	x	x	x	p	p	r	t	t	r	x		
GJ	x	x	g	x	x	g	x	x	r	t	p	p	t	r	w	t	h	x	h	w	t	x	t	h	x	h	w	t	r	
GK	r	t	p	r	t	x	x	p	t	m	m	r	t	r	r	w	t	h	x	w	h	x	x	t	h	t	w	h	x	r
GM	t	w	h	t	w	x	h	x	r	m	m	t	r	r	r	t	r	p	x	t	x	p	x	r	p	r	t	x	p	t
HJ	w	x	x	h	t	h	t	w	x	x	h	t	h	w	t	x	p	t	r	p	t	r	x	x	x	g	x	x	t	
HL	t	x	r	p	r	t	p	x	r	x	p	r	t	x	p	t	t	m	m	r	r	r	w	w	t	h	x	x	h	w
HN	w	x	t	h	t	w	x	h	t	x	h	t	w	h	x	r	r	m	m	t	r	r	t	t	r	p	x	p	x	t
IM	x	w	w	t	h	t	h	x	t	t	r	p	r	p	x	r	t	x	p	r	p	x	t	r	t	m	t	r	r	x
IO	x	t	t	r	p	r	x	p	w	w	t	h	t	x	h	t	w	x	h	t	x	h	r	t	r	m	m	r	r	x
JH	t	r	x	t	r	p	p	x	x	r	t	r	p	x	p	w	t	w	t	h	h	x	t	t	w	x	h	h	x	m
JN	w	t	x	w	t	h	x	h	x	t	w	t	h	h	x	t	r	t	r	p	x	p	r	r	t	x	p	x	p	m
KL	m	r	r	r	m	r	t	t	r	r	r	m	r	t	t	x	p	t	r	t	p	x	x	x	x	x	g	g	t	
KM	r	t	x	r	t	p	p	x	r	m	m	r	r	t	t	r	x	p	t	p	x	p	r	x	r	t	p	x	r	
KN	t	r	p	t	r	x	x	p	p	r	t	r	x	p	x	t	r	t	r	x	p	x	r	t	p	x	p	x	m	
KO	r	m	r	m	r	t	t	x	t	r	t	p	x	p	x	x	x	x	x	g	g	r	p	t	r	t	p	x	r	
LM	r	x	t	p	t	r	p	x	m	r	r	m	t	t	r	r	r	m	m	t	t	x	t	p	t	r	x	p	x	
LN	t	p	r	x	r	t	x	p	r	p	x	r	t	p	x	r	r	m	m	r	t	t	t	t	r	x	p	x	p	t
LO	r	r	m	r	r	m	t	t	t	x	p	t	r	x	p	p	x	r	t	r	x	p	t	x	r	t	r	x	p	x
MN	x	x	x	x	x	g	g	r	r	t	p	t	p	x	m	m	r	r	r	t	t	r	m	r	r	r	t	t	r	
MO	p	t	t	r	x	r	p	x	t	t	r	x	x	p	r	t	p	x	r	x	p	r	r	r	m	m	t	t	p	
NO	x	r	r	t	p	t	x	p	x	x	x	x	g	g	r	t	r	t	p	x	p	m	r	m	r	t	t	r		
	AB	AE	AG	AH	AI	AJ	AM	AN	BC	BF	BH	BI	BJ	BN	BO	CD	CF	CG	CI	CJ	CK	CO	DE	DF	DG	DH	DJ	DK	DL	EF

EG EH EI EL EM FI FH FL FO GI GJ GK GM HJ HL HH IM IO JK JN KL KM KN KO LM LN LO MN MO NO																														
t	x	x	r	t	a	x	r	t	x	x	r	t	w	a	x	t	w	m	r	t	r	r	t	r	a	p	a	AB		
h	g	h	p	h	x	x	t	r	w	x	t	w	a	x	x	w	t	r	t	r	t	r	m	a	p	r	a	t	r	AE
h	x	w	t	w	t	r	m	m	h	g	p	h	a	r	t	w	t	a	x	r	a	p	r	t	r	m	a	t	r	AG
x	g	x	x	p	p	t	r	t	x	r	t	h	p	h	t	r	t	w	r	r	t	m	p	x	r	a	r	t	AH	
w	x	h	t	w	p	p	r	t	h	x	t	w	t	r	t	h	p	r	t	m	t	r	r	t	r	r	a	x	p	AI
a	x	t	r	t	r	t	m	m	a	g	a	x	h	t	w	t	r	p	h	r	p	a	r	r	t	m	x	r	t	AJ
w	x	w	a	h	t	r	r	r	w	a	x	h	t	p	x	h	a	p	x	t	p	a	t	p	a	t	g	p	x	AM
t	x	t	p	a	r	t	r	r	t	a	p	a	w	x	h	a	p	x	h	t	a	p	t	a	p	t	g	a	p	AN
r	r	t	m	m	w	a	t	w	a	a	t	r	a	r	t	t	w	a	x	r	r	p	a	m	r	t	r	t	a	BO
r	x	a	t	r	h	g	p	h	t	r	m	m	a	x	a	t	w	r	t	r	m	r	t	r	p	a	r	t	a	BF
r	p	p	t	r	x	g	a	x	r	t	m	m	h	p	h	r	t	t	w	r	m	t	r	r	a	p	t	r	a	BH
t	p	p	r	t	h	a	t	w	p	p	r	t	t	r	t	p	h	r	t	m	r	r	t	r	r	t	p	a	x	BI
r	t	r	m	m	t	a	r	t	p	p	t	r	h	t	w	r	t	p	h	r	r	a	p	m	t	r	t	r	a	BJ
m	t	r	r	r	t	x	p	x	r	t	r	r	w	a	h	p	a	x	h	t	t	p	a	t	p	a	p	a	g	BN
m	r	t	r	r	w	a	x	h	t	r	r	r	t	p	a	x	h	p	a	t	t	a	p	t	a	p	a	p	g	BO
a	t	p	t	r	a	x	a	x	a	w	w	t	a	t	r	r	t	w	t	a	t	t	a	r	r	p	m	r	r	CD
p	t	a	t	r	h	p	p	h	a	t	t	r	p	t	r	t	w	t	r	p	r	r	a	r	r	x	m	t	t	CF
p	r	x	t	r	x	r	r	t	g	h	h	p	t	m	m	a	x	w	t	t	a	t	x	r	m	r	r	p	r	CG
x	r	p	r	t	h	t	t	w	g	a	x	x	r	m	m	p	h	t	r	r	p	r	x	r	m	t	r	a	t	CI
t	r	r	m	m	t	p	r	t	x	h	w	t	p	r	t	r	t	h	p	t	t	a	x	m	r	r	r	r	p	CJ
t	m	r	r	r	t	r	p	a	x	w	h	a	t	r	r	p	a	h	a	p	p	p	g	t	t	a	t	a	x	CK
r	m	t	r	r	w	t	a	h	a	t	x	p	r	r	r	a	h	a	p	a	x	a	g	t	t	p	t	p	p	CO
g	h	p	h	p	x	w	w	t	x	a	x	a	x	w	t	t	r	t	r	a	p	r	r	a	t	t	r	r	m	DE
x	w	a	w	t	p	h	h	p	p	t	t	r	a	w	t	r	t	t	r	a	r	r	p	t	t	x	m	r	r	DF
g	a	a	x	p	t	t	r	p	h	h	p	a	t	r	t	r	w	t	a	x	t	t	p	r	r	r	r	m	DC	
a	h	t	w	t	t	h	w	t	r	a	t	r	g	h	p	m	m	a	x	a	r	p	r	t	a	t	r	m	r	DH
x	a	r	t	r	r	a	t	r	t	h	w	t	g	a	x	t	m	h	p	a	t	a	t	r	p	r	r	m	r	DJ
a	t	r	a	p	r	t	a	p	t	w	h	a	a	x	p	r	r	h	a	g	p	p	p	a	x	a	t	t	t	DK
a	w	t	h	a	t	w	h	a	r	t	a	p	a	h	a	r	r	a	p	g	a	a	x	p	p	p	t	t	t	DL
p	h	g	h	p	g	h	h	p	a	r	r	t	t	w	t	a	x	m	m	t	r	m	r	a	t	a	r	p	r	EF
p	h	p	h	a	t	t	r	h	p	p	h	p	t	r	w	t	t	r	p	a	r	r	a	r	r	t	t	m	EG	
p	p	h	p	a	h	w	t	t	p	r	t	p	h	p	t	r	r	t	t	r	r	m	a	x	a	t	p	r	r	EH
h	p	p	h	g	a	x	a	h	t	t	w	r	t	r	h	p	m	m	r	t	m	r	a	r	p	t	a	r	EI	
p	h	p	g	a	w	h	a	t	r	p	a	t	h	a	x	p	r	r	p	a	t	t	g	p	p	a	x	t	EL	
h	p	h	g	a	t	a	p	w	t	x	h	r	a	p	h	a	r	r	a	p	t	t	g	a	p	p	t	EN		
x	x	g	a	a	p	p	h	p	r	r	t	r	t	r	p	h	m	m	r	r	m	t	p	r	a	r	a	t	FI	
t	h	a	w	t	p	h	p	r	r	m	m	p	h	p	r	t	r	t	t	m	r	r	t	x	x	r	r	p	FH	
t	w	a	h	a	p	h	g	r	m	r	r	t	h	a	p	a	r	r	p	t	t	x	p	p	g	t	a	a	FL	
r	t	a	x	p	h	p	g	t	m	r	r	r	a	p	a	h	r	r	a	t	t	p	a	x	g	t	p	p	FO	
h	t	h	t	w	p	r	r	t	p	p	h	r	m	m	h	p	t	r	r	a	x	r	p	t	m	r	t	a	r	GI
p	p	t	r	t	r	r	m	m	p	h	p	p	r	t	m	r	h	p	t	a	x	t	r	r	m	p	r	r	GJ	
p	r	t	p	a	r	m	r	r	p	h	g	t	r	r	a	p	h	a	p	g	p	p	a	t	t	a	x	t	GK	
h	t	w	a	h	t	m	r	r	h	p	g	r	r	r	h	a	x	p	a	g	a	x	p	t	t	p	p	t	GM	
p	p	r	t	r	r	p	t	r	r	p	t	r	p	h	m	m	p	h	p	r	a	r	r	a	r	t	m	t	NJ	
t	h	t	h	a	t	h	h	x	m	r	r	r	p	g	r	r	p	a	p	t	x	t	p	g	p	a	t	x	HL	
r	p	r	a	p	r	p	a	p	m	t	r	r	h	g	r	r	a	h	a	t	p	t	a	g	a	p	t	p	HN	
w	t	h	a	h	p	r	p	a	h	m	a	h	m	r	r	g	r	r	t	p	t	x	p	t	x	p	g	a	IM	
t	r	p	p	a	h	t	a	h	p	r	p	a	m	r	r	g	r	r	t	x	t	p	a	t	p	a	g	p	IO	
t	r	m	r	r	m	r	r	r	t	h	h	x	p	p	a	r	r	g	p	p	g	p	t	a	t	x	t	x	JK	
r	t	m	r	r	m	t	r	r	r	p	x	p	h	a	h	r	r	g	a	x	g	a	t	p	t	p	t	p	JN	
p	t	r	p	a	r	t	p	a	r	t	p	x	p	p	a	t	t	p	a	h	h	h	h	h	w	w	w	KL		
x	r	t	x	p	r	m	t	t	x	a	g	g	r	t	t	p	a	p	a	h	h	h	w	w	h	h	w	KN		
r	r	m	t	t	m	r	t	t	r	a	p	a	x	a	p	t	t	g	g	h	h	h	w	h	w	h	h	KN		
r	m	r	t	t	t	r	x	p	p	t	p	x	r	t	t	x	p	p	a	h	h	h	w	w	h	h	h	KO		
x	a	x	g	g	p	t	p	a	t	r	x	p	r	p	a	p	a	t	t	h	h	w	w	h	h	h	w	LM		
r	a	r	p	a	r	a	p	x	m	r	t	t	x	g	g	t	t	x	p	h	w	h	w	h	h	h	w	LN		
r	t	p	p	a	x	x	g	g	r	m	t	t	r	p	a	x	p	t	t	h	w	w	h	h	h	w	h	LO		
t	p	t	x	p	r	r	t	t	t	p	a	p	t	x	p	p	a	x	p	w	h	h	w	h	h	w	h	h	MN	
t	r	a	x	p	x	r	a	p	x	r	x	p	m	t	t	g	g	t	t	w	h	w	h	h	h	h	h	ND		
m	r	r	t	t	t	p	x	p	r	r	t	t	t	x	p	a	p	x	p	w	w	h	h	w	h	h	h	h	NO	
EG EH EI EL EM FI FH FL FO GI GJ GK GM HJ HL HH IM IO JK JN KL KM KN KO LM LN LO MN MO NO																														



402.

ON A SINGULARITY OF SURFACES.

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A SURFACE having a nodal line has in general on this nodal line points where the two tangent planes coincide, or as I propose to term them "pinch-points." Thus, if the nodal line be the curve of complete intersection of any two surfaces $P=0$, $Q=0$, then the equation of the general surface having this curve for a nodal line is $(a, b, c \chi P, Q)^2=0$ (where a, b, c are any functions of the coordinates), and the pinch-points are given as the intersections of the nodal line $P=0$, $Q=0$ with the surface $ac-b^2=0$. Consider the case where the nodal curve is a curve of partial intersection represented by the equations $\| P, Q, R \| = 0$, or say by the equations $p=0, q=0, r=0$ (viz. p, q, r denote the functions $QR-Q'R, RP-R'P, PQ-P'Q$ respectively), and consequently we have identically

$$(P, Q, R \chi p, q, r) = 0,$$

$$(P', Q', R' \chi p, q, r) = 0,$$

or what is the same thing, (λ, μ) being arbitrary,

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' \chi p, q, r) = 0.$$

The general surface having the curve in question for its nodal line is represented by the equation

$$(a, b, c, f, g, h \chi p, q, r)^2 = 0,$$

(where (a, b, c, f, g, h) are any functions of the coordinates), and it is easy to see that the condition for a pinch-point is the same as that which (considering p, q, r as coordinates and all the other quantities as constants), expresses that the line

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' \chi p, q, r) = 0,$$

touches the conic

$$(a, b, c, f, g, h)(p, q, r)^2 = 0,$$

viz. A, B, C, F, G, H being the inverse coefficients, $A = bc - f^2$, &c., this condition is

$$(A, B, C, F, G, H)(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R')^2 = 0,$$

or what is the same thing, the pinch-points are given as the common intersections of the nodal line $p = 0, q = 0, r = 0$ with each of the three surfaces

$$(A, B, C, F, G, H)(P, Q, R)^2 = 0,$$

$$(A, B, C, F, G, H)(P, Q, R)(P', Q', R') = 0,$$

$$(A, B, C, F, G, H)(P', Q', R')^2 = 0,$$

these last three equations in fact, adding only a single relation to the relations expressed by the equations

$$p = 0, q = 0, r = 0.$$

If the functions P, Q, R, P', Q', R' are linear functions of the coordinates, then the curve $(p = 0, q = 0, r = 0)$ is a cubic curve in space, or skew cubic; and if moreover (a, b, c, f, g, h) are constants, then the equation

$$(a, b, c, f, g, h)(p, q, r)^2 = 0,$$

belongs to a quartic surface having the skew cubic for a nodal line: this surface is (it may be observed) a ruled surface, or scroll. With a view to ulterior investigations, I propose to study the theory of the pinch-points in regard to this particular surface; and to simplify as much as possible, I fix the coordinates as follows:

Considering the skew cubic as given, let any point O on the cubic be taken for the origin; let $x = 0$ be the equation of the osculating plane at O ; $y = 0$ that of any other plane through the tangent line at O ; $z = 0$, that of any other plane through O , not passing through the tangent line; and $w = 0$ that of a fourth plane; then the equation of the cubic will be

$$\begin{vmatrix} x & y & z \\ y & z & w \end{vmatrix} = 0,$$

or what is the same thing, the values of p, q, r are $yw - z^2, zy - xw$, and $xz - y^2$ respectively. And conversely, the cubic being thus represented, the point $(x = 0, y = 0, z = 0)$ may be considered as standing for any point whatever on the skew cubic; the osculating plane at this point being $x = 0$, and the tangent line being $x = 0, y = 0$. For the purpose of the present investigations, we may without loss of generality write $w = 1$; and for convenience I shall do this; the values of p, q, r thus become $y - z^2, yz - x, xz - y^2$, and the equation of the surface is

$$(a, b, c, f, g, h)(y - z^2, yz - x, xz - y^2)^2 = 0.$$